

An Independent project report

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[Smart Clinic Booking System]

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Authorship Declaration

Except where reference is made in the references, this report contains no material published elsewhere or extracted in whole or in part from a dissertation or report presented by me for another degree or diploma.

No other person's work has been used without due acknowledgement in the content of the report.

Signature: _____WAI_____ Date:_____03 Apr 2026_____

[WAI YAN HTET]

Acknowledgement

To start it off, I would like to express my gratitude and thank my supervisor, Dr. Saleh Abdulalem Ali Mohammed for his guidance throughout the duration of this project. He has dedicated his time to mentor me which has greatly contributed to the completion of this project. The advice and feedback given has ultimately enhanced both my understanding and improved the quality of my work.

Other than that, I am also grateful to my friends who had previously taken the CD209 - Managing & Implementing Business Project for sharing their valuable advice and experiences. They have provided encouragement and helpful guidance which greatly helped me in completing this project.

Abstract

The Smart Clinic Booking System is a web-based application designed to make the process of scheduling medical appointments easier, faster, and more efficient. In many clinics, appointment booking is still done manually, which can cause long waiting times, double bookings and poor record management. This system helps solve these problems by providing an online platform where patients can easily register, log in, view available doctors, and book appointments based on their preferred date and time. Patients can also check, update or cancel their appointments anytime and gives them more flexibility and convenience. At the same time, the system allows administrators to manage doctor information, approve or reject appointment requests and monitor daily schedules. This helps reduce administrative workload and improves the overall management of clinic operations. The system also includes basic validation to prevent booking conflicts and ensure that appointment data is accurate and well-organized. The system was developed using web technologies such as PHP, MySQL, HTML, CSS, and Bootstrap. A structured design approach was followed using UML diagrams, including use case diagrams, activity diagrams and sequence diagrams to clearly show how the system works. The system was tested through functional testing and user feedback to evaluate its usability, reliability and performance. The results show that the system is user-friendly, easy to understand and efficient in managing clinic appointments. Most users were satisfied with the system features and found it helpful in reducing time and effort. Overall, the Smart Clinic Booking System provides a simple, convenient and well-organized solution for both patients and clinic administrators and it has the potential to be further improved for real-world use.

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Chapter 1: Introduction

1.10 Background

The rapid growth of web technologies has changed how service industries manage their operations. Many organizations are moving from manual systems to digital platforms to improve efficiency, accessibility and service quality. In the healthcare sector, appointment management is very important for ensuring smooth clinic operations and reducing patient waiting time.

Traditionally, many small and medium-sized clinics still use manual appointment booking methods such as phone calls, walk-in registration and paper-based scheduling. These traditional systems often cause long waiting times, double booking problems, poor data management and ineffective communication between patients and clinic administrators.

With the development of web-based systems, clinics can now use online appointment booking platforms that allow patients to schedule appointments anytime and anywhere. A web-based clinic booking system can improve efficiency, reduce administrative work, increase data accuracy and provide a better experience for patients.

This project proposes the development of a Smart Clinic Booking System. The system allows patients to register, log in, book appointments and manage their schedules online. At the same time, administrators can manage doctors, approve appointments and monitor daily schedules more effectively.

1.2 Problem Statement

Many clinics still rely on manual or semi-manual appointment systems. These systems cause several issues which are without proper validation, multiple patients may book the same doctor at the same time and leading to operational confusion. Inefficient appointment management Administrators need to manually check schedules and inform doctors about daily appointments, which increases workload and risk of errors. If these issues are not addressed, clinics may face poor service quality, longer waiting times, and decreased patient satisfaction. This project will develop a web-based clinic booking system using PHP, MySQL, and Bootstrap. The system will implement proper validation mechanisms to prevent booking conflicts and provide real-time appointment management.

Justification

Implementing a web-based booking system will:

- Improve clinic operational efficiency
- Reduce scheduling conflicts
- Enhance patient convenience
- Provide better data organization and security

1.3 Aim

This project aims to develop a web-based Smart Clinic Booking System to improve appointment scheduling efficiency and enhance service quality in clinic operations.

1.4 Objectives

1. To study the existing manual and online clinic appointment systems.
2. To develop a functional clinic booking system using PHP and MySQL.
3. To test and evaluate the system functionality through user testing.

1.5 Justification

The development of the Smart Clinic Booking System is supported by both academic and practical reasons. From an academic perspective, this project combines knowledge from different subjects such as Database Systems, Web Development, System Analysis and Design and Software Project Management. It allows the use of theoretical concepts like database design (ERD), system Modeling (Use Case Diagram and Activity Diagram) and the system development lifecycle (SDLC) in a real-world system. From a practical perspective, many small and medium clinics still use manual or partly digital appointment systems. These systems often have problems such as double booking, inefficient scheduling and poor record management. By developing a web-based booking system, clinic operations can be automated and improved, reducing administrative work and increasing service quality. This project is also useful for personal development because it improves technical skills in PHP, MySQL, system validation, and user interface design. It also helps to develop problem-solving skills and gives practical experience in building a full-stack web application.Furthermore, the system can be used in real-world situations with better security and proper hosting setup, making it suitable for future expansion beyond academic use.

1.6 Scope

The scope of this project defines the functional boundaries and limitations of the Smart Clinic Booking System.

Target Users

The system will support two types of users:

1. Administrator
2. Patient (Client)

System Functional Scope

Patient Functions:

- Register account with email validation
- Login and logout
- Book appointment with available doctors
- View personal appointment history
- Cancel approved or pending appointments
- Receive registration confirmation email

Administrator Functions:

- Secure login access
- Add, edit, and delete doctor information
- View all appointments
- Approve or decline appointment requests
- View today's approved schedule
- Prevent deletion of patients with active appointments

Technical Scope

- Web-based system
- Developed using PHP, MySQL, HTML, CSS, Bootstrap
- Localhost deployment using XAMPP
- Role-based access control (Admin and Patient)
- Appointment validation mechanism to prevent: double booking, booking unavailable time slots and same patient booking multiple appointments at same time

Out of Scope

The following features are not included in this project:

- Online payment gateway integration
- SMS notification system
- Multi-branch clinic management
- Real-time doctor availability tracking
- Mobile application (Android/iOS native app)
- Advanced medical record management system

This project focuses specifically on appointment booking and schedule management as a prototype system.

1.7 Constraints and Assumptions

Constraints

- The project must be completed within the allocated academic timeline, limiting the development to core system functionalities only.
- The system is developed using PHP and MySQL within a local XAMPP environment.
- Advanced security features such as SSL encryption and multi-factor authentication are not implemented as this is a prototype system.
- The system is designed for a single clinic branch and does not support multi-branch operations.

Assumptions

- Users (administrator and patients) have basic knowledge of using web-based systems.
- Internet connectivity is stable during system usage.
- Appointment time slots are managed and maintained by the administrator.
- Users provide accurate and valid information during registration.

1.8 Major Risk

Identified Risks

- Technical Implementation Risk – Errors in coding or database configuration may cause system malfunction or data inconsistency.
- Time Management Risk – Delays in development or testing may affect project completion within the academic timeline.
- Data Integrity Risk – Improper validation may lead to double booking or incorrect appointment records.
- System Security Risk – As the system is a prototype, limited security measures may expose vulnerabilities if deployed publicly.
-

1.9 External Bodies Involved

SDG 3: Good Health and Well-Being

SDG 3 aims to ensure healthy lives and promote well-being for all at all ages. This project supports SDG 3 by developing a Smart Clinic Booking System that improves appointment scheduling efficiency, reduces waiting time and enhances access to healthcare services. By digitalizing clinic operations, the system contributes to better healthcare service management and patient satisfaction.

SDG 4: Quality Education

SDG 4 ensures inclusive and quality education while promoting lifelong learning opportunities. This project supports SDG 4 by providing hands-on experience in web development, database design and system implementation. Through the development of this system, practical and technical skills are enhanced in software development and problem-solving.

SDG 9: Industry, Innovation and Infrastructure

SDG 9 focuses on building resilient infrastructure, promoting sustainable, industrialization, and fostering innovation. This project supports SDG 9 by developing a web-based clinic management system that enhances digital infrastructure within small healthcare organizations and promotes innovation through the adoption of technology driven solutions.

1.10 Project Planning

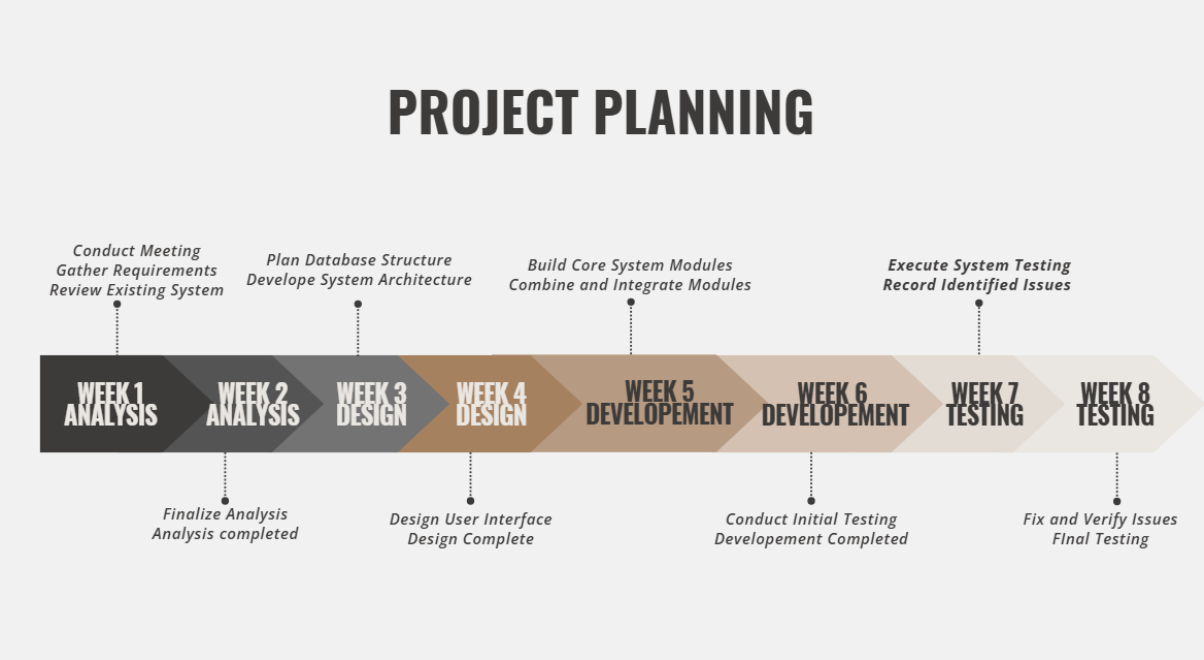


Figure 1.10.1 Project Planning

Chapter 2: Literature Review

2.1 Introduction

The rapid development of information technology has greatly changed the healthcare industry. Many healthcare organizations are moving from manual and paper-based systems to digital platforms to improve efficiency, data accuracy and patient satisfaction. One important development is the use of web-based appointment booking systems which help reduce waiting time, lower administrative workload and improve access to services.

Traditional clinic appointment systems usually depend on phone calls, walk-in registration or manual record-keeping. These methods can easily lead to human errors, appointment conflicts, data loss and poor communication between patients and administrators. As the demand for convenient healthcare services increases, clinics need reliable and user-friendly systems that allow patients to book appointments online and manage their schedules more efficiently.

This literature review looks at previous studies on web-based appointment systems, healthcare management information systems and scheduling methods. The aim is to identify the strengths and weaknesses of existing systems and explain the need to develop a Smart Clinic Booking System with proper validation and role-based access control.

2.2 Web-Based Appointment Scheduling Systems

Web-based appointment scheduling systems allow patients to book consultations online without visiting the clinic. According to Zhang et al. (2018), online appointment systems improve hospital efficiency by giving real-time access to available time slots and reducing overcrowding in healthcare facilities [1]. These systems also improve transparency and allow patients to choose their preferred dates and doctors more easily.

In addition, Gupta and Denton (2008) stated that structured scheduling systems help improve the use of healthcare resources and reduce appointment delays [2]. Efficient scheduling reduces doctor's idle time and ensures that patients receive treatment on time. However, many early scheduling systems did not include proper web integration which limited their accessibility and scalability.

Despite these benefits, some studies show that web-based appointment systems still face problems such as double booking, overlapping schedules and weak validation systems. Without real-time database checking, more than one patient may book the same time slot can cause inefficiency in clinic operations. Therefore, modern systems need to include real-time validation to ensure accurate and reliable appointment scheduling.

2.3 Healthcare Management Information Systems (HMIS)

Healthcare Management Information Systems (HMIS) combine different clinic operations such as patient registration, appointment scheduling and administrative tasks into one digital system. According to Wager, Lee, and Glaser (2017), digital healthcare systems improve data accuracy, support better decision-making and reduce the use of paper in healthcare organizations [3]. These systems help improve resource management and overall service quality.

In addition, centralized systems make it easier to track data and generate reports. Clinic administrators can monitor daily schedules, review patient records and create reports more efficiently. However, the implementation of healthcare information systems may face challenges such as security risks, data privacy issues and resistance from users, especially in small clinics with limited resources [3].

These challenges show the importance of designing a system that is simple, secure and easy to use especially for small and medium-sized clinics.

2.4 Appointment Conflict and Validation Mechanisms

One of the main challenges in healthcare scheduling systems is preventing appointment conflicts and double booking. Cayirli and Veral (2003) explained that poor scheduling methods can cause patient congestion, longer waiting times and underuse of healthcare resources [4]. Good scheduling requires proper time-slot allocation and effective conflict detection.

Modern systems solve these problems by using database rules and real-time validation checks. For example, before confirming an appointment, the system checks whether the selected doctor and time slot are already booked. These validation methods improve system reliability and reduce errors in clinic operations.

In addition, role-based access control is important for maintaining system security and proper functionality. By allowing only authorized users to perform administrative tasks (such as approving appointments or managing doctors), the system prevents unauthorized access and improves overall control.

2.5 Summary of Literature Review

The reviewed literature shows that web-based appointment systems greatly improve efficiency, accessibility and overall management in healthcare institutions. However, some limitations still exist such as scheduling conflicts, weak validation systems and security issues.

Based on the findings, it is evident that an effective clinic booking system must include:

- Real-time appointment validation
- Role-based access control
- Centralized data management

- User-friendly interface

Therefore, the proposed Smart Clinic Booking System aims to address these limitations by implementing conflict prevention mechanisms, structured user roles and efficient appointment management features.

No	Title	Year	Focus	Limitations
1	Online Appointment Scheduling in Healthcare	2018	Web-based booking efficiency	Limited to hospital case study
2	Appointment Scheduling in Healthcare	2008	Scheduling optimization models	No web-based validation
3	Health Care Information Systems	2017	Digital healthcare management	High implementation cost
4	Outpatient Scheduling Policies	2003	Conflict prevention and scheduling policies	Traditional scheduling models

Figure (2.1) Summary of Literature Review

Chapter 3: Research Methods

3.1 Introduction

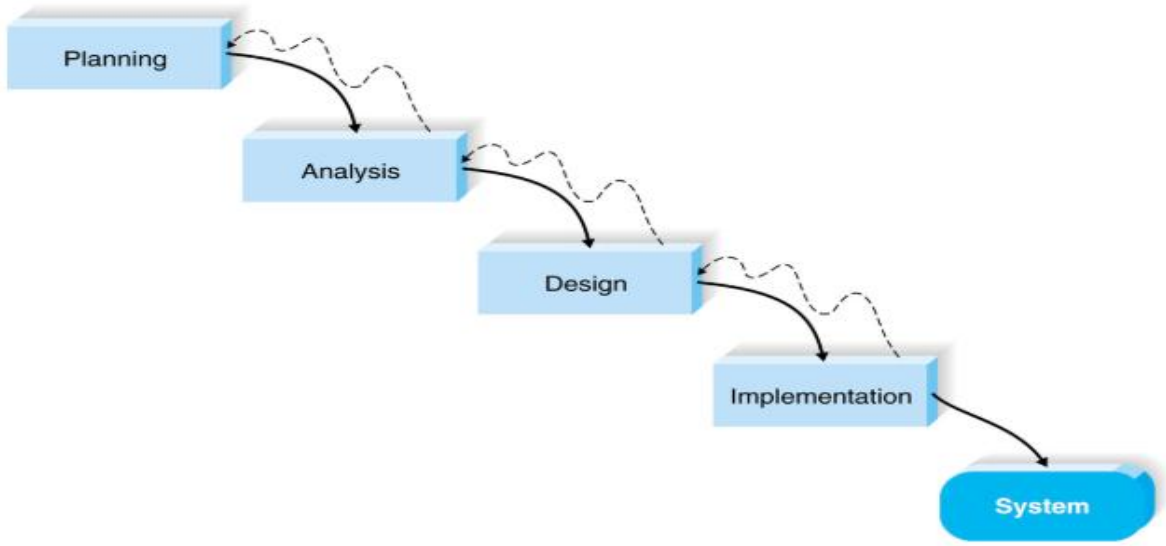
Research methods provide a structured way to collect and analyze information in order to achieve the objectives of a study. In this project, methods such as literature review and system analysis are used to understand existing clinic booking systems, their limitations and user requirements.

The information collected from previous studies helps guide the design and development of the proposed Smart Clinic Booking System. By using a suitable system development method, the project follows a clear process for planning, designing, implementing and testing the system effectively.

3.2 Methodology Used

Methodology Chosen: Waterfall Development Model

The Waterfall Development Model is a linear and structured approach where each phase is completed before moving to the next. It is suitable for this project because the system requirements are clearly defined, and the development process follows a step-by-step sequence.



Planning Phase

In this phase, the project scope, objectives, and system requirements are identified. The main goal is to develop a Smart Clinic Booking System that allows patients to book appointments online and helps administrators manage doctors and schedules efficiently.

Analysis Phase

During the analysis phase, existing manual and online booking systems are studied to identify their weaknesses. Problems such as double booking, lack of validation, and inefficient scheduling are examined. The requirements of both patients and administrators are also clearly defined.

Design Phase

In this phase, the system structure and database design are created. The system includes features such as user registration, login, appointment booking, doctor management and schedule management. The user interface is also designed to ensure it is simple and easy to use.

Implementation Phase

The system is developed using PHP, MySQL, HTML, CSS, and Bootstrap. Important functions such as booking validation, role-based access control and appointment management are implemented. Each part of the system is coded and combined into a complete system.

Testing Phase

In this phase, the system is tested to make sure all functions work properly. This includes testing the login system, appointment booking, validation checks and user interactions. Errors and bugs are found and fixed to improve the system's reliability.

System Deployment

Finally, the system is deployed in a local environment using XAMPP. The completed system is ready for demonstration and can be improved further for real-world use.

3.3 System Requirement

Functional Requirements

User registration and login system

Role-based access control (Admin and Patient)

Doctor management (add, edit, delete doctor information)

Appointment booking system for patients

Appointment approval and cancellation by admin

Patient appointment management (view and cancel appointments)

Display doctor availability (day and time)

Prevent double booking and scheduling conflicts

View daily appointment schedule for administrators

Non-Functional Requirements

Only registered users can log in, ensuring system security

The system should provide fast response time during booking and data retrieval

The system must ensure accurate data handling and prevent booking conflicts

The system should be reliable and stable during usage

The system should have a user-friendly interface with easy navigation

3.4 Summary

This chapter explains the methodology and system requirements for developing the Smart Clinic Booking System. The project uses the Waterfall Development Methodology which follows a clear and step-by-step process where each phase is completed before moving to the next.

The system is developed using web-based technologies such as PHP, MySQL, HTML, CSS, and Bootstrap. It requires a local server environment like XAMPP to run properly. The defined functional and non-functional requirements help guide the development process to ensure the system is efficient, reliable and easy to use.

Chapter 4: System Implementation and Development

4.1 Introduction

This chapter presents the results of developing the Smart Clinic Booking System. It shows how the system was designed and built to meet the project objectives and requirements. The chapter also explains the overall system structure using UML diagrams to show how users interact with the system.

In addition, this chapter describes the implementation of key features such as user registration, login, appointment booking, doctor management and appointment approval. The results show that the system improves clinic operations by providing an efficient and user-friendly online booking platform.

4.2 System Architecture and Design

The Smart Clinic Booking System is designed as a web-based application that allows interaction between two main users: Administrator and Patient. The system uses a client-server structure where users access the system through a web browser, and data is stored and managed in a MySQL database.

The Administrator is responsible for managing doctors, approving or rejecting appointments, and viewing daily schedules. The Patient can register, log in, book appointments, view their bookings and cancel them when needed. The system includes proper validation to prevent appointment conflicts and uses role-based access control to ensure security and proper system functionality.

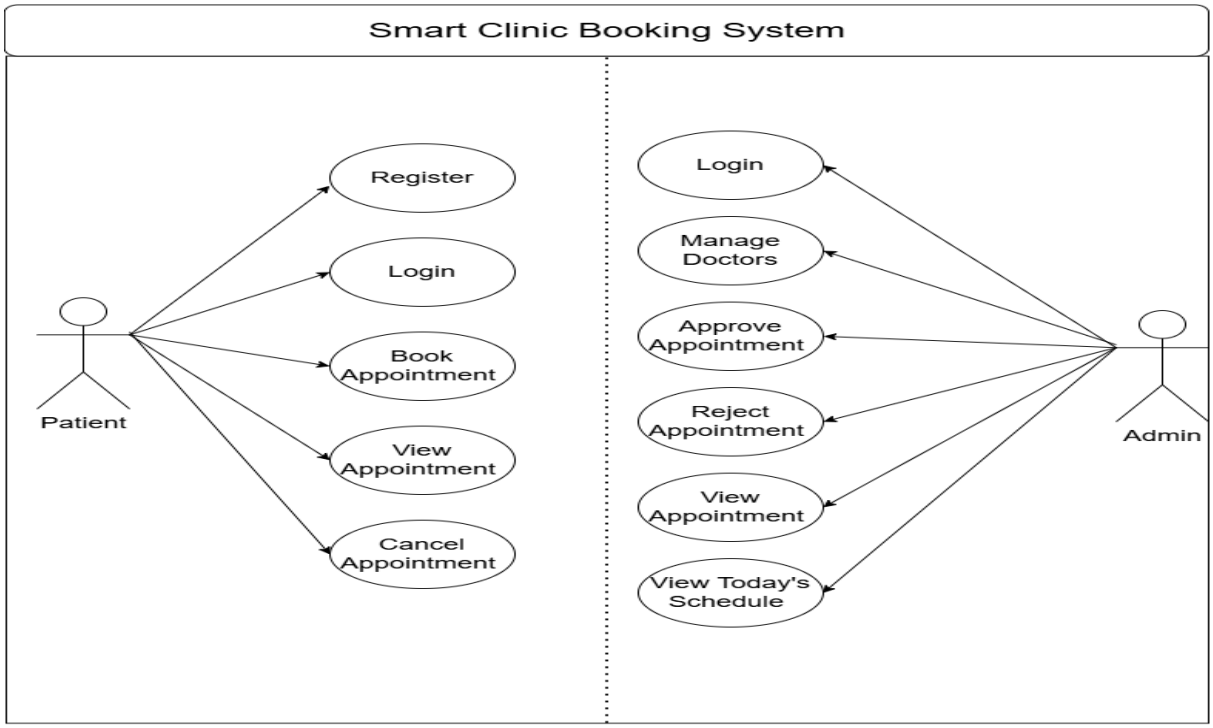


Figure 4.1 Use Case Diagram

Figure 4.1 Use Case Diagram: The Use Case Diagram shows the interaction between two main users: Patient and Administrator. The Patient can register, log in, book appointments, view their bookings and cancel them. The Administrator is responsible for managing doctors, viewing and controlling appointments, approving or cancelling bookings and monitoring the daily schedule. This diagram shows the overall functions of the system and how users interact with the Smart Clinic Booking System.

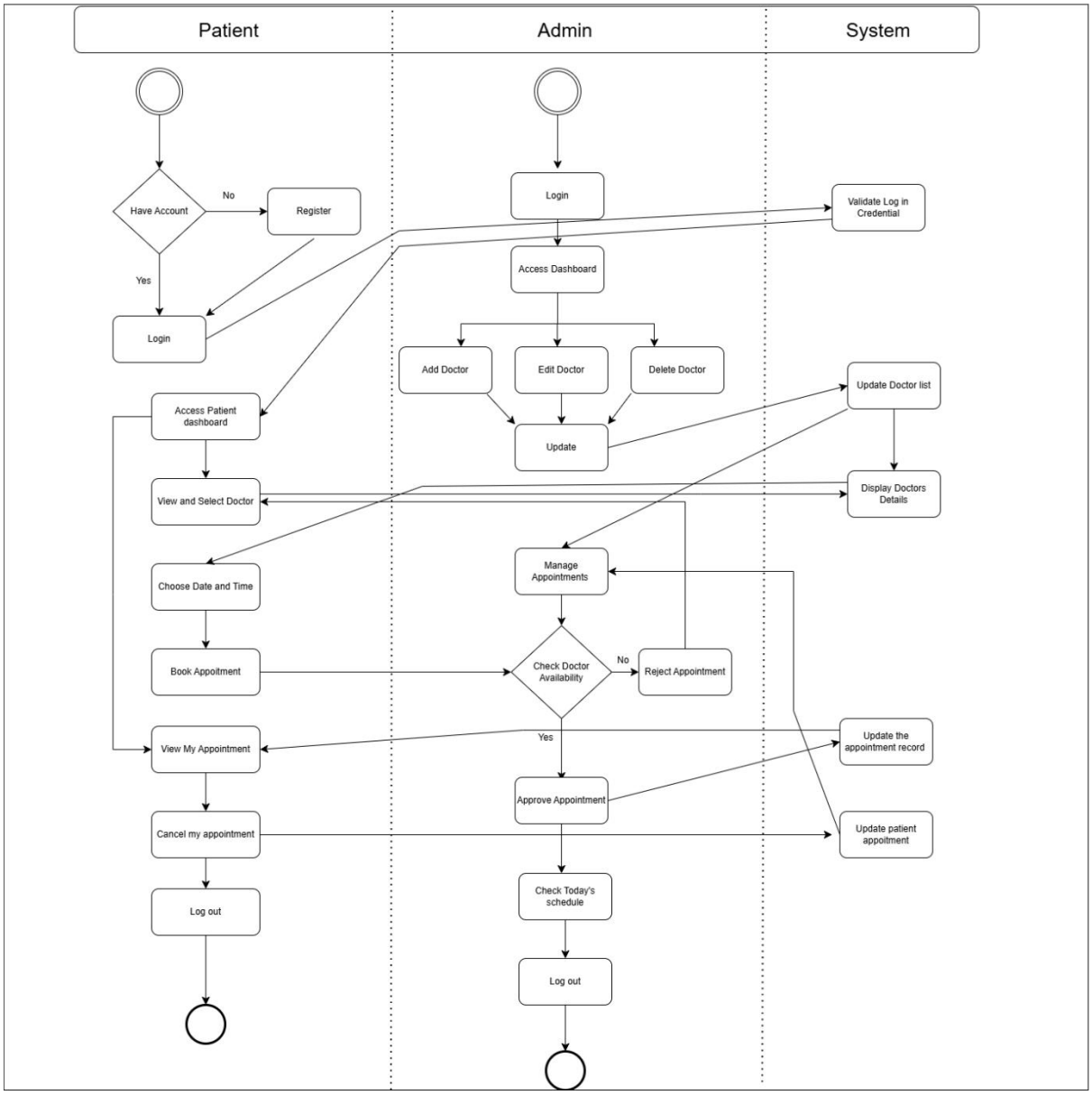


Figure 4.2 Activity Diagram

Figure 4.2 Activity Diagram: The Activity Diagram shows the workflow of the Smart Clinic Booking System with three main entities: Patient, Administrator, and System. The Patient can register, log in, view doctors, choose available time slots, book appointments, view their bookings and cancel them. The Administrator logs in to manage doctors by adding, editing or deleting their information and handles appointment requests by approving or rejecting them based on availability. The System is responsible for validating login details, checking doctor availability, displaying doctor information and updating appointment records. This diagram shows the overall process flow and the interaction between users and the system to ensure efficient appointment management.

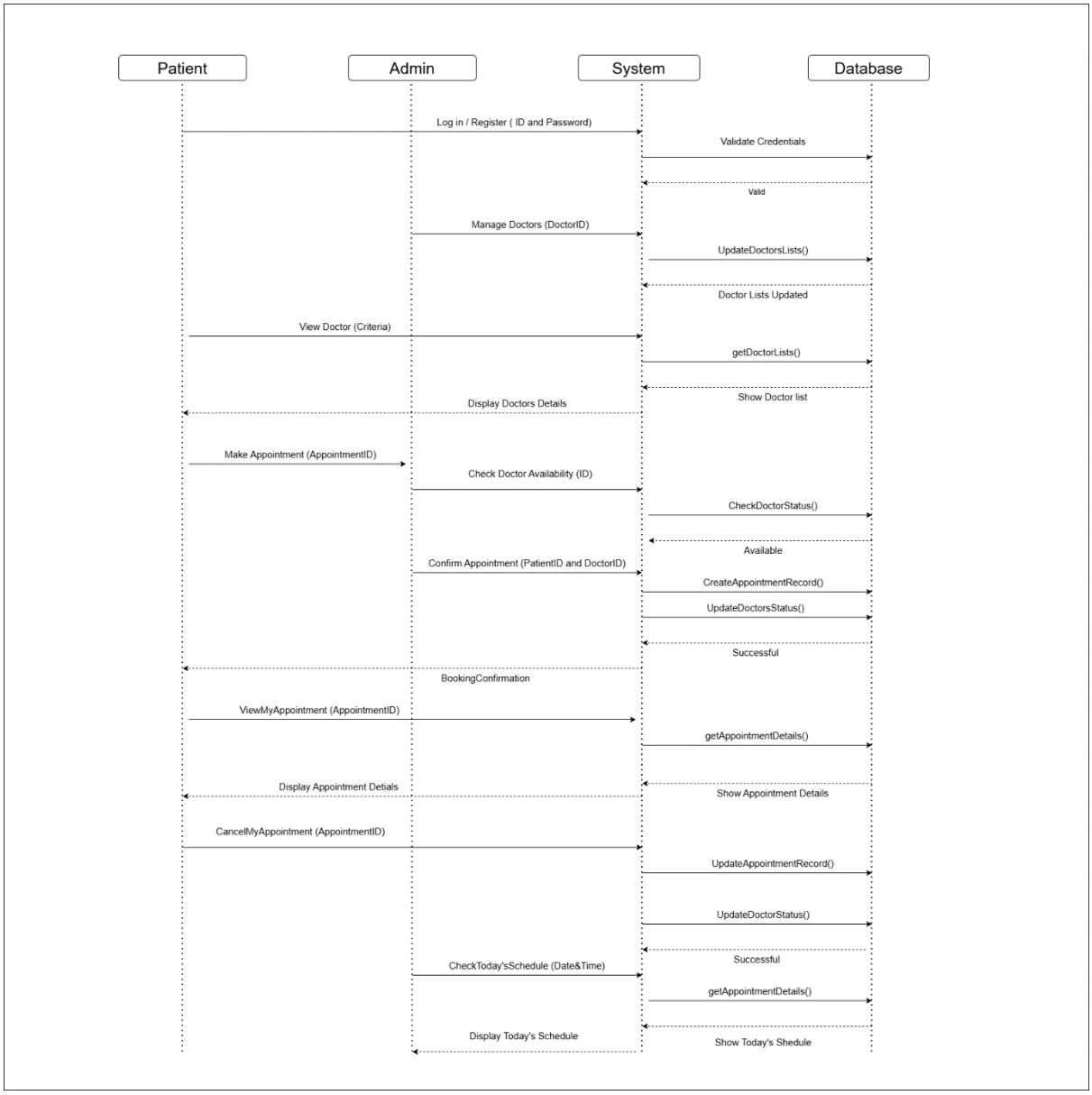


Figure 4.3 Sequence Diagram

Figure 4.3 Sequence Diagram: The Sequence Diagram shows the interaction between Patient, Administrator, System and Database in the Smart Clinic Booking System. The process starts when the patient or administrator logs in or registers and the system checks the login details with the database.

The administrator manages doctor information, and the system updates the doctor records in the database. The patient can view available doctors, and the system retrieves and displays the information. When the patient books an appointment, the system checks the doctor's availability in the database. If the time slot is available, the system confirms the booking and creates the appointment record and updates the doctor's schedule.

The patient can also view and cancel appointments where the system retrieves or updates the records in the database. In addition, the administrator can check the daily schedule and the system displays the related appointment details. This diagram shows the sequence of

interactions and data flow between users, the system and the database to ensure efficient appointment management.

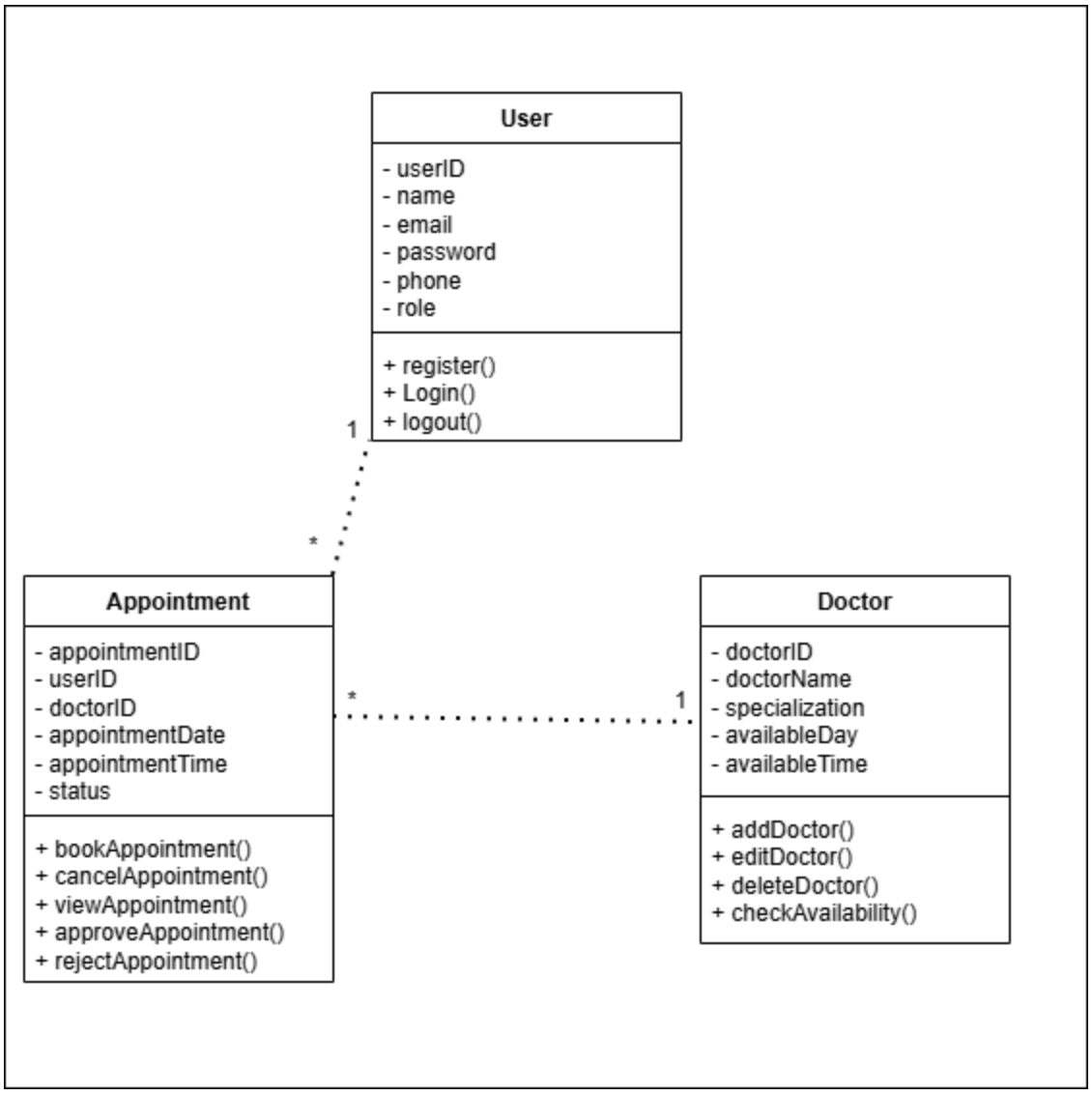


Figure 4.4 Class Diagram

Figure 4.4 Class Diagram: The Class Diagram includes three main classes: User, Doctor, and Appointment. The User class handles login and authentication. The Doctor class manages doctor information while the Appointment class handles booking and status updates. The relationships show that one user can have many appointments and each doctor can manage multiple appointments. This diagram shows the structure of the system and how different parts are connected and work together.

4.3 System Development

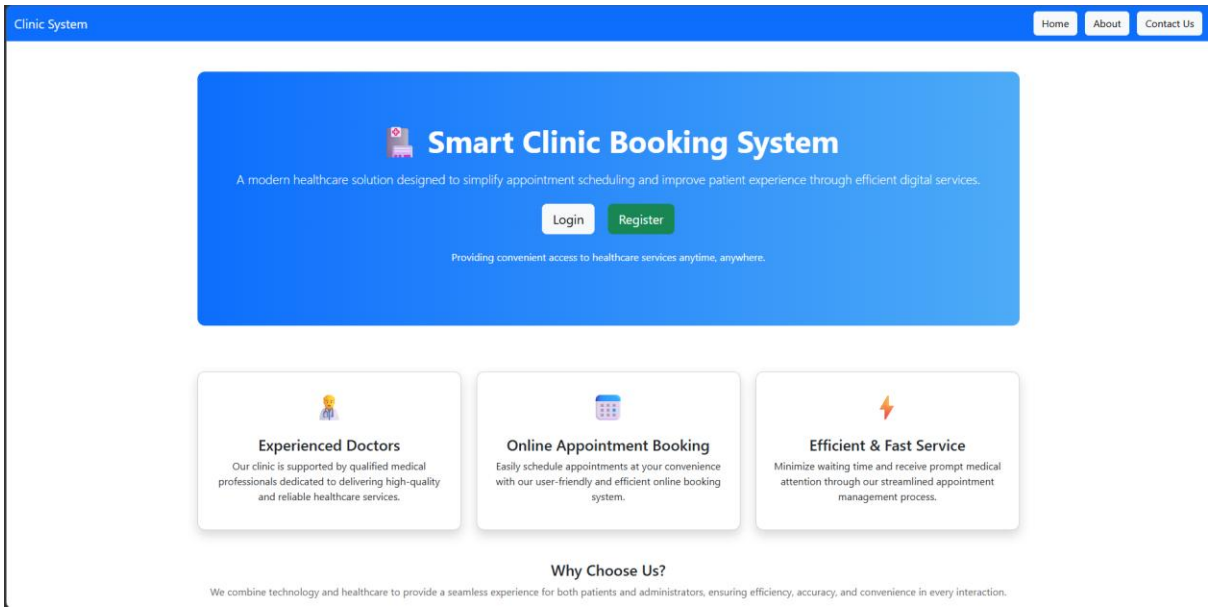


Figure 4.3.1 Home Page

The homepage presents the Smart Clinic Booking System with navigation, login and registration options and key features such as online booking and efficient service. It provides a simple and user-friendly interface for users to access the system.

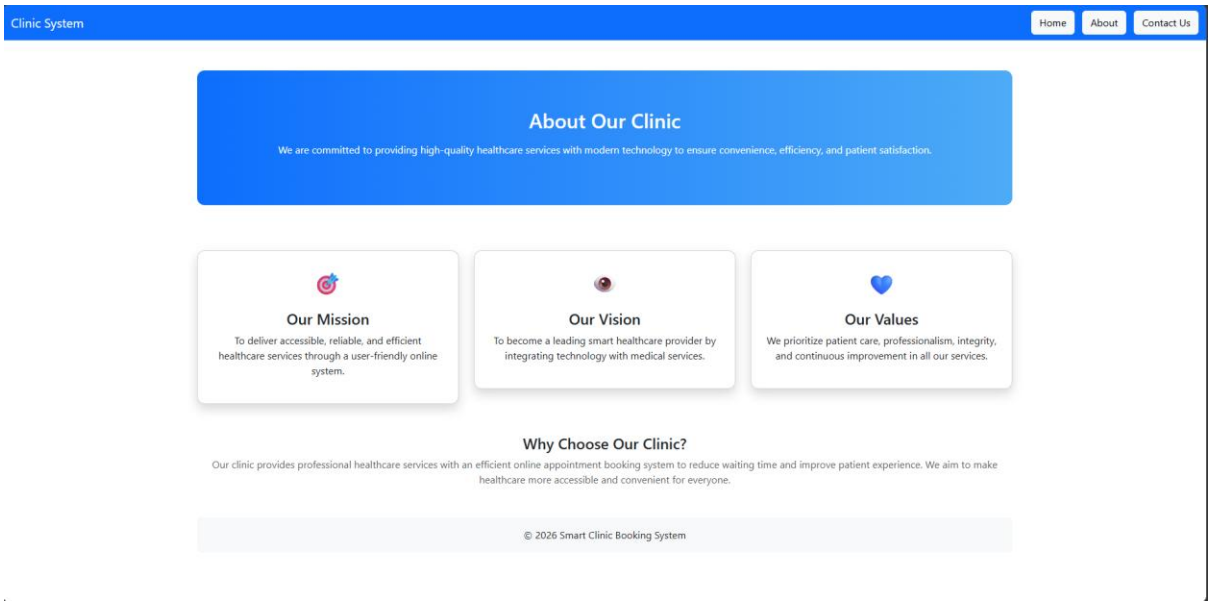


Figure 4.3.2 About Page

The About page presents information about the clinic, including its mission, vision and values. It helps users understand the purpose and services of the Smart Clinic Booking System

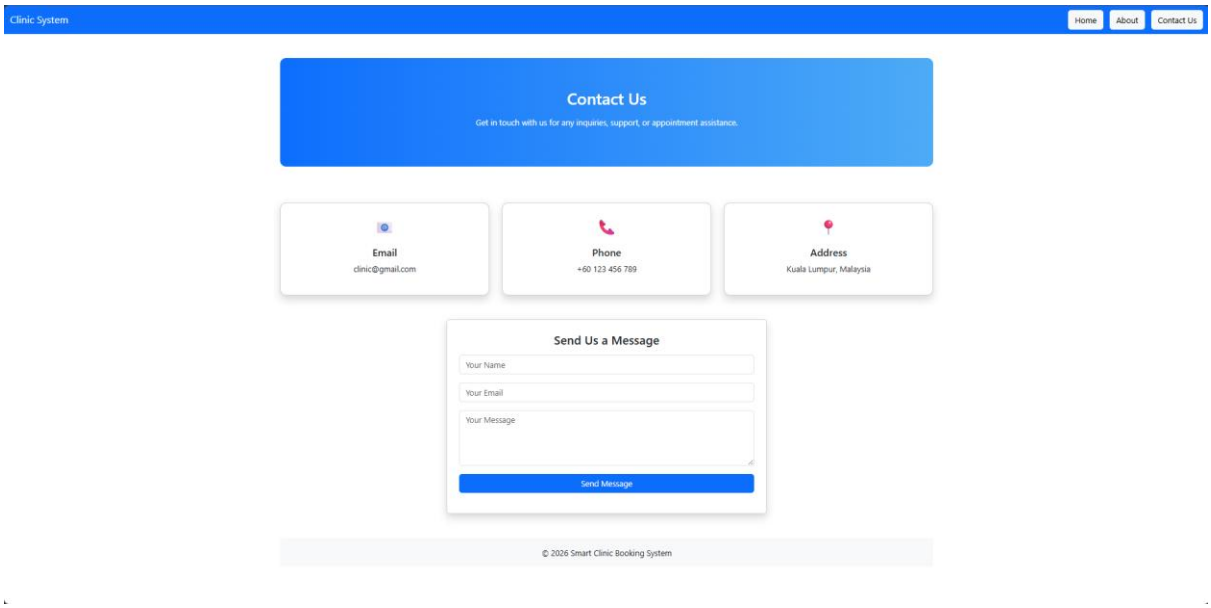


Figure 4.3.3 Contact Us Page

The Contact page displays the clinic's contact details and provides a form for users to send messages or inquiries.

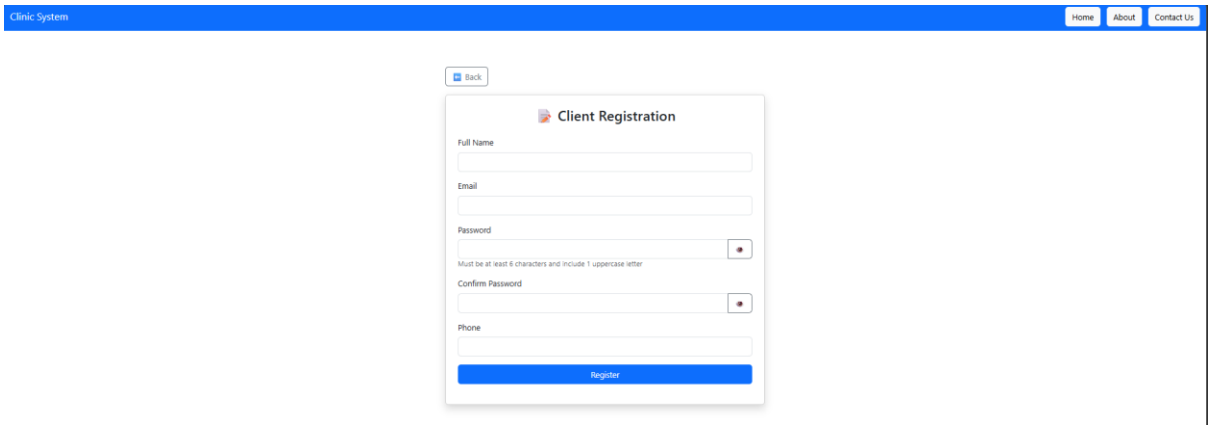


Figure 4.3.4 Registration Page

The Registration page allows users to create an account by entering their details, with validation to ensure accuracy and security.

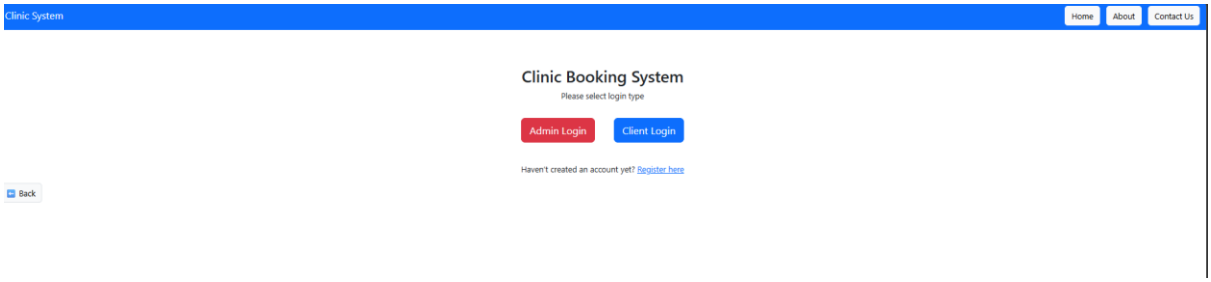


Figure 4.3.5 Login Selection Page

The page allows users to select either Admin or Client login and provides a registration link for new users.

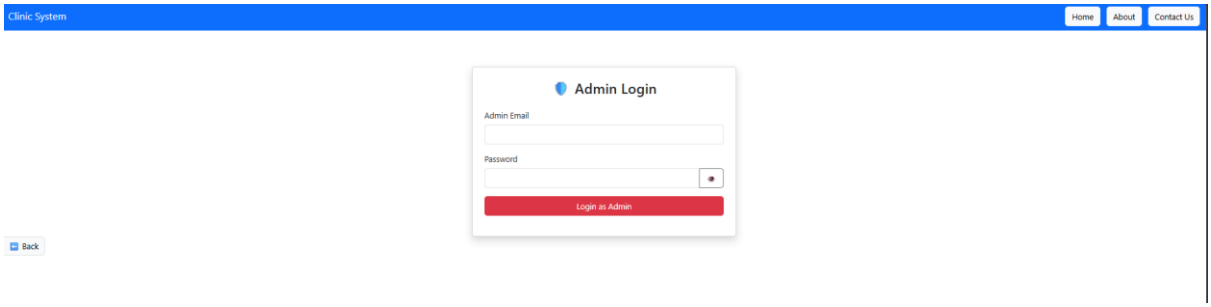


Figure 4.3.6 Admin Login Page

The Admin Login page allows administrators to enter their credentials and securely access the system.

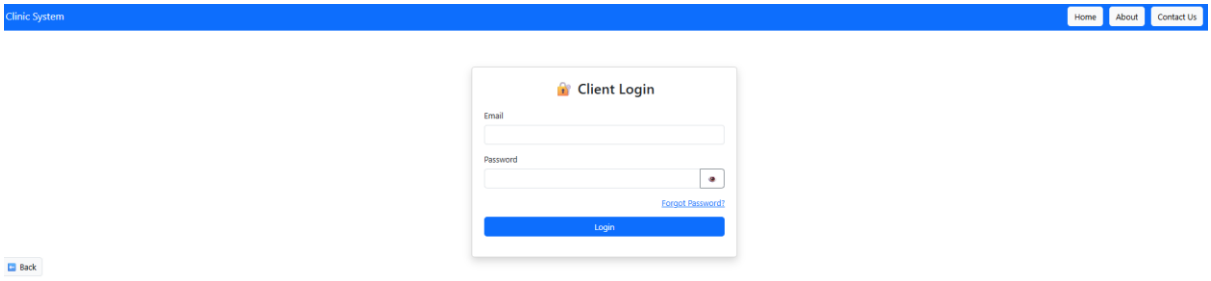


Figure 4.3.7 Client login Page

The Client Login page allows users to enter their credentials and access the system with options for password visibility and password recovery.

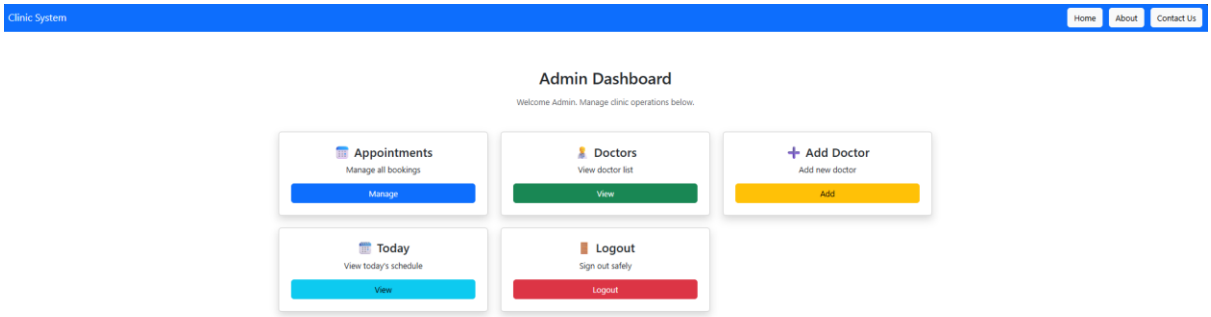


Figure 4.3.8 Admin Dashboard Page

The Admin Dashboard provides access to core functions such as managing appointments, doctors and schedules by allowing administrators to control the system efficiently.

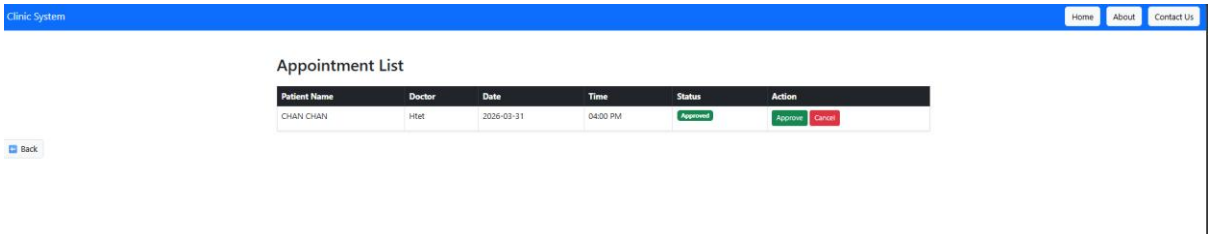


Figure 4.3.9 Appointment Page

The page displays appointment details and allows administrators to approve or cancel bookings for efficient management.

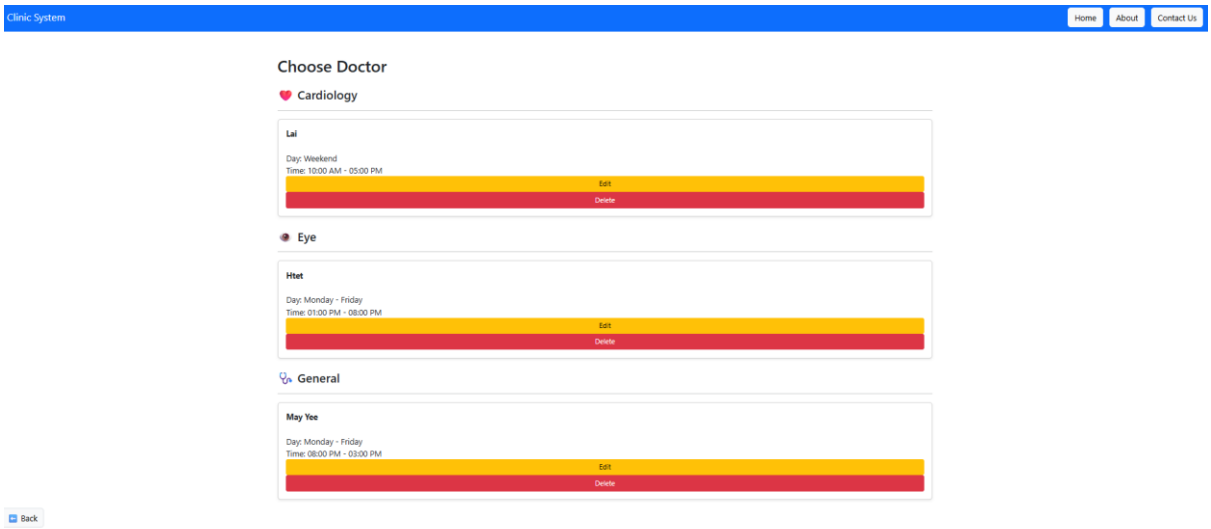


Figure 4.3.10 Doctor List Page

The page displays doctors by specialization with their availability and allows administrators to edit or delete doctor details.



Figure 4.3.11 Edit Doctor Page

The page allows administrators to update doctor details such as specialization and availability.



Figure 4.3.12 Add Doctor Page

The page allows administrators to add new doctor details, including specialization and availability for system management.

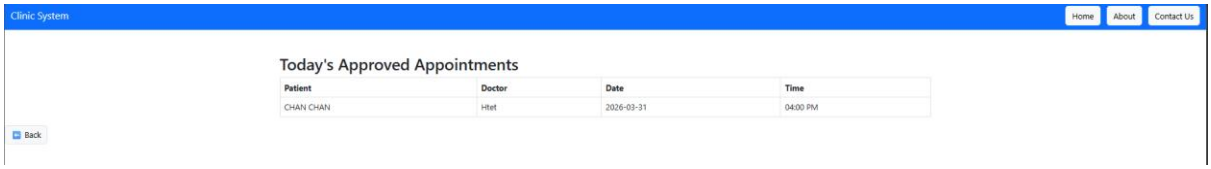


Figure 4.3.13 Appointment Schedule Page

The page displays all approved appointments for the current day by allowing administrators to monitor the daily schedule efficiently.

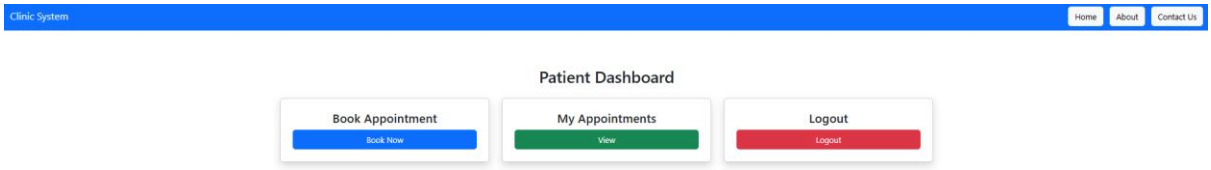


Figure 4.3.14 Patient Dashboard Page

The Patient Dashboard allows users to book appointments, view their bookings and log out through a simple and user-friendly interface.

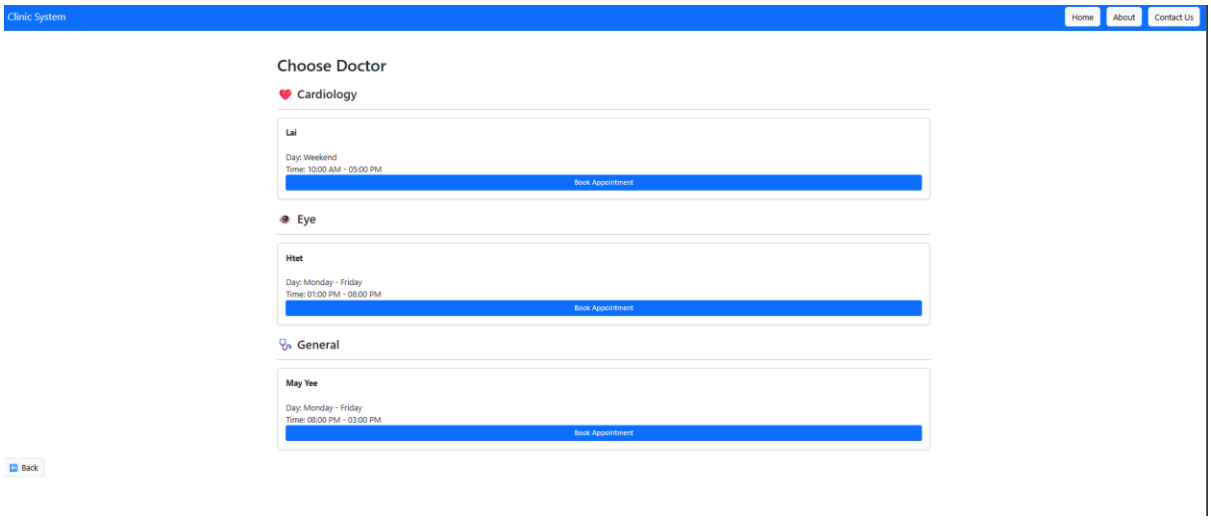


Figure 4.3.15 View to Make Appointment Page

The page allows patients to view doctors by specialization and select a doctor to book an appointment.

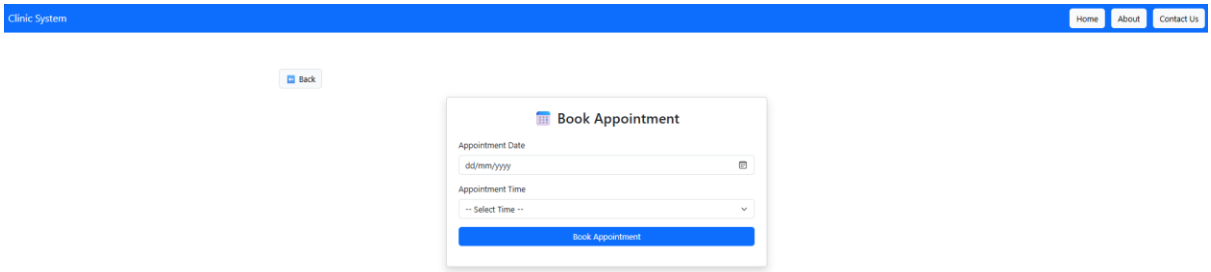


Figure 4.3.16 Book Appointment Page

The page allows patients to select a date and time to book an appointment based on doctor availability.

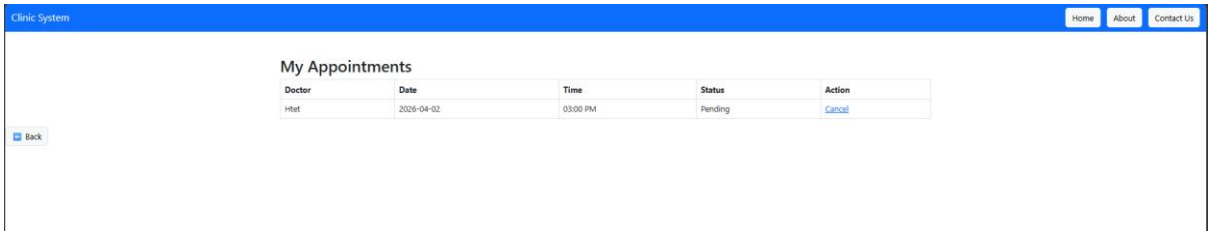


Figure 4.3.17 Check My Appointment Page

The page displays a patient's appointments and allows them to view details and cancel bookings.

4.4 Summary

The Smart Clinic Booking System is developed using a client–server architecture where users interact through a web-based interface built with HTML, CSS and Bootstrap. PHP is used for server-side processing and MySQL is used for data storage. The system allows smooth communication between the interface and the database to manage user, doctor and appointment information efficiently.

The system successfully meets its objectives by providing a functional and user-friendly platform. Patients can register, log in, book appointments and manage their bookings while administrators can manage doctors, approve or reject appointments and monitor daily schedules. Overall, the system improves efficiency, reduces manual work and enhances the quality of healthcare services.

Chapter 5: Discussion & Conclusion

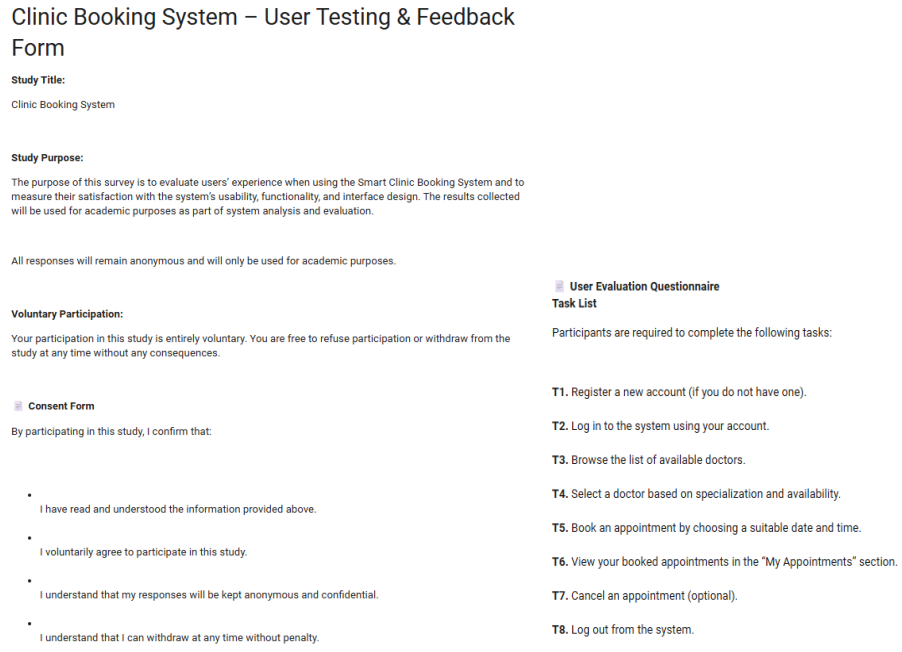
5.1 Introduction

This chapter evaluates the performance and usability of the Smart Clinic Booking System based on the results from system testing and user feedback. The collected responses are analyzed to measure the system's effectiveness, efficiency and user satisfaction. In addition, the achievements and limitations faced during the development of this project are also discussed.

5.2 Evaluation and Discussion

A total of 10 participants were involved in testing the Smart Clinic Booking System. The evaluation was conducted using a structured questionnaire after users completed a series of tasks including registration, login, browsing doctors, booking appointments and managing their bookings. The feedback collected provides insights into the system's usability, functionality and overall user experience.

User Acceptance Testing



Clinic Booking System – User Testing & Feedback Form

Study Title:

Clinic Booking System

Study Purpose:

The purpose of this survey is to evaluate users' experience when using the Smart Clinic Booking System and to measure their satisfaction with the system's usability, functionality, and interface design. The results collected will be used for academic purposes as part of system analysis and evaluation.

All responses will remain anonymous and will only be used for academic purposes.

Voluntary Participation:

Your participation in this study is entirely voluntary. You are free to refuse participation or withdraw from the study at any time without any consequences.

Consent Form

By participating in this study, I confirm that:

- I have read and understood the information provided above.
- I voluntarily agree to participate in this study.
- I understand that my responses will be kept anonymous and confidential.
- I understand that I can withdraw at any time without penalty.

User Evaluation Questionnaire

Task List

Participants are required to complete the following tasks:

T1. Register a new account (if you do not have one).

T2. Log in to the system using your account.

T3. Browse the list of available doctors.

T4. Select a doctor based on specialization and availability.

T5. Book an appointment by choosing a suitable date and time.

T6. View your booked appointments in the "My Appointments" section.

T7. Cancel an appointment (optional).

T8. Log out from the system.

Figure 5.2.1 User Testing and Feedback Form

The form collects user feedback through consent, task-based testing, and questionnaires to evaluate system usability and performance.

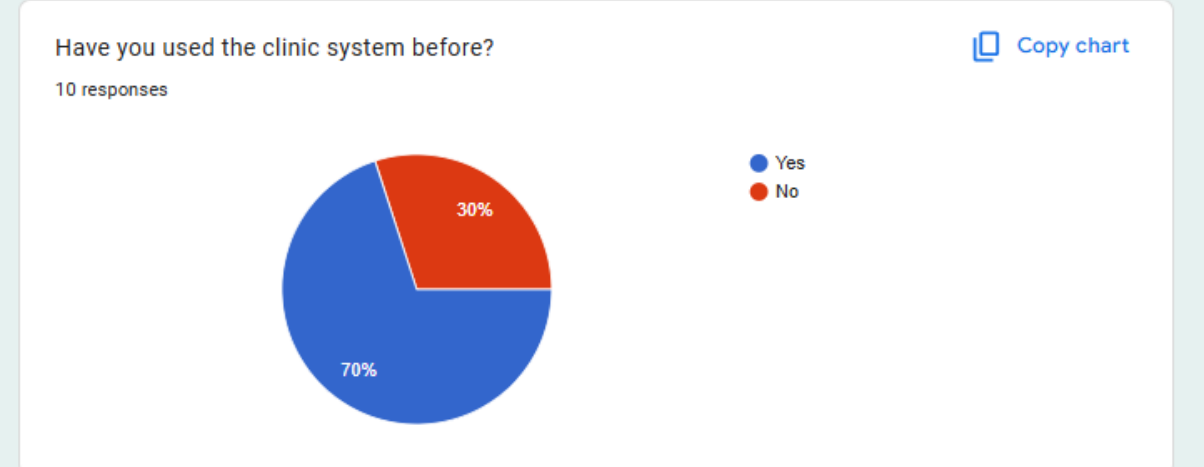


Figure 5.2.2 Experience with Clinic System

Most participants (70%) have prior experience with clinic systems, while 30% are new users.

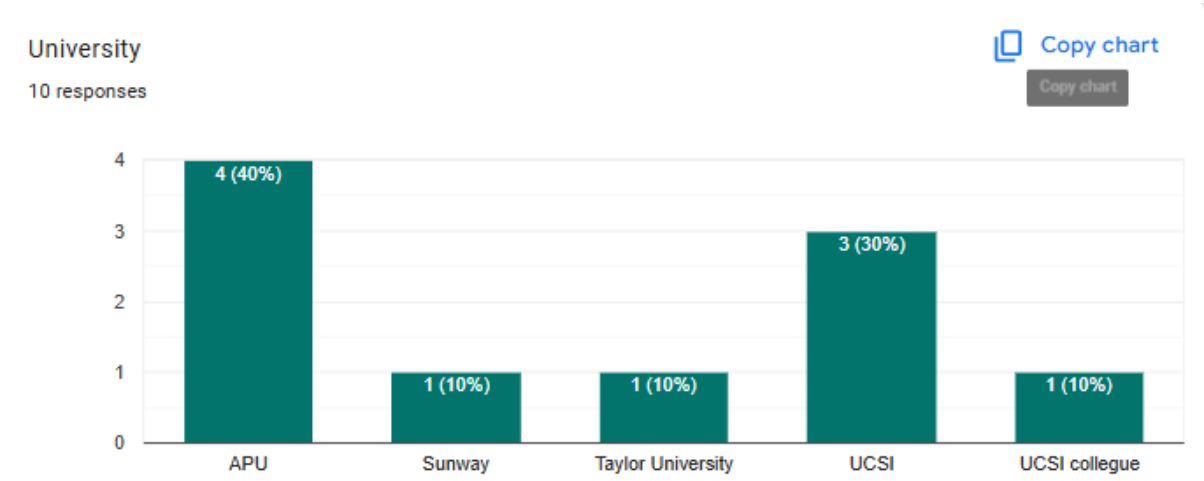


Figure 5.2.3 Participants from University

Most respondents are from APU (40%) and UCSI (30%), with smaller contributions (10% each) from other universities.

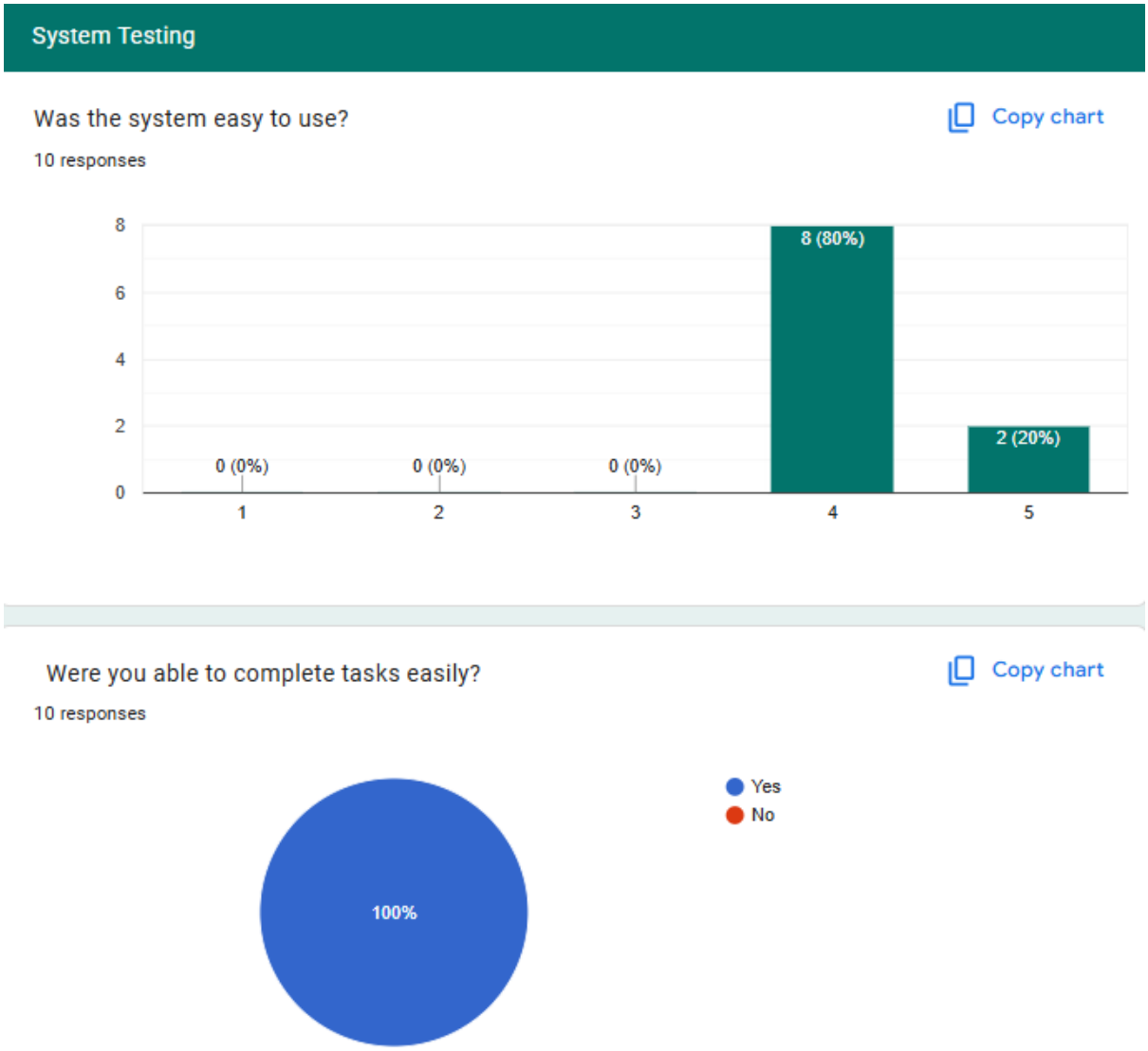


Figure 5.2.4 Usability and Task Completion

Most users rated the system as easy to use (80% rated 4, 20% rated 5), and all participants (100%) were able to complete tasks successfully.

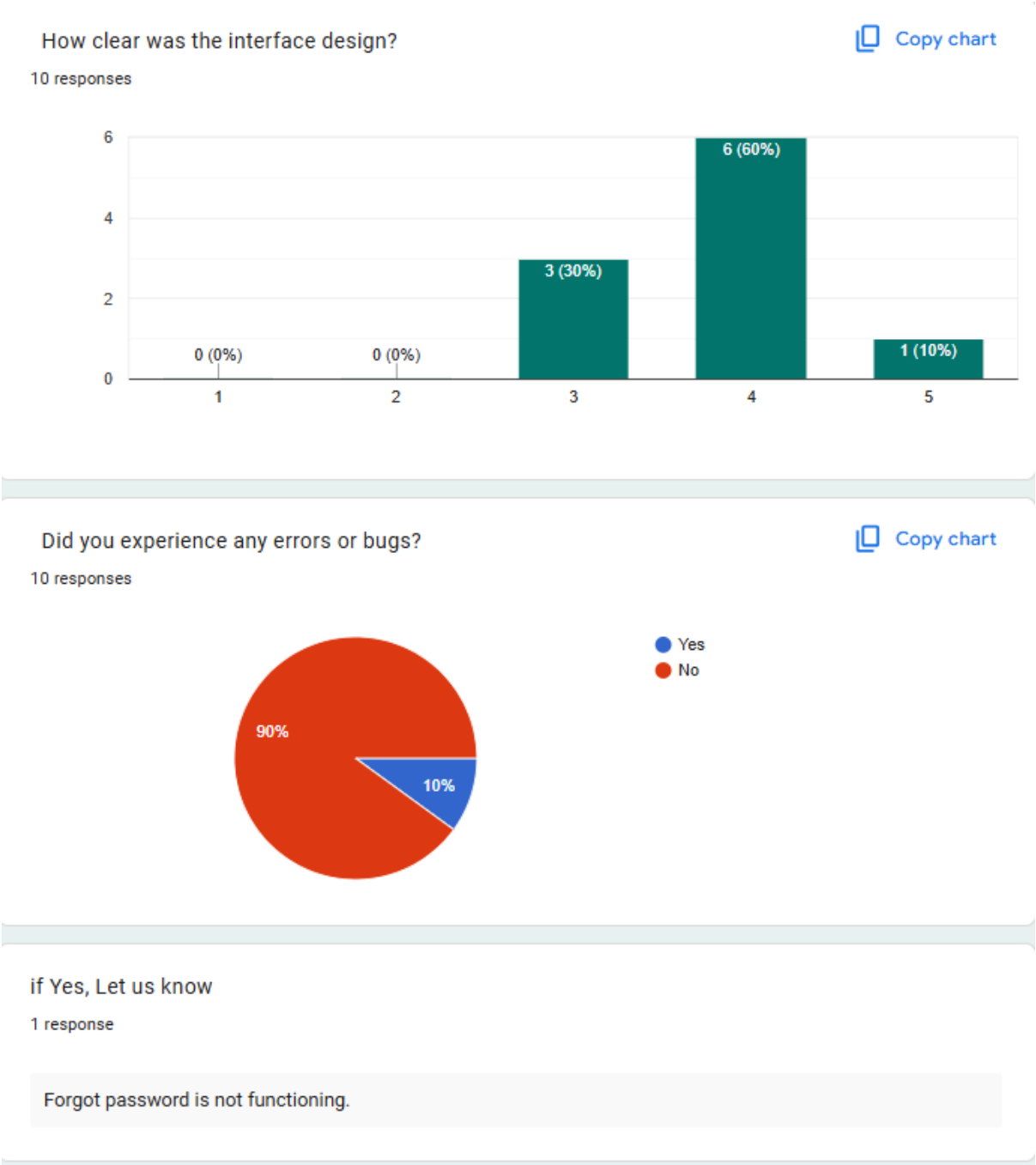


Figure 5.2.5 System Design and System Error

This figure shows users' evaluation of the interface design and system reliability. For interface clarity, 60% of the respondents gave a rating of 4, 30% gave a rating of 3 and 10% gave a rating of 5. This indicates that most users found the interface clear and easy to understand. No users gave a rating below 3 which shows a generally positive opinion of the design.

For system errors, 90% of the respondents said they did not face any problems, while 10% reported a small issue. The problem mentioned was related to the "Forgot Password" function not working properly. Overall, the system shows good reliability with only a few minor issues.

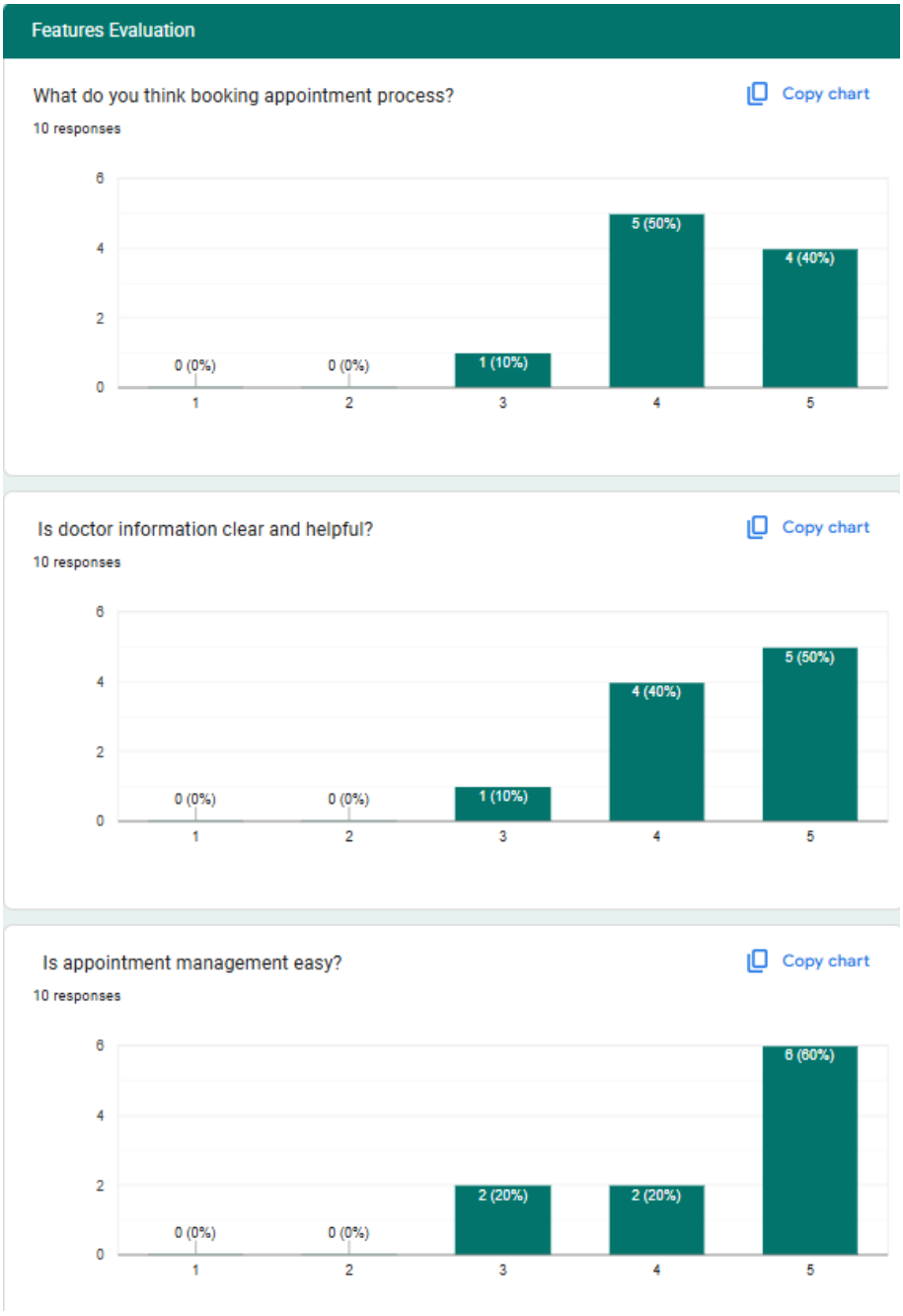


Figure 5.2.6 Features Evaluation

This figure shows users' evaluation of the system features. Most respondents gave high ratings for the appointment booking process, mainly scores of 4 and 5 which means the process is efficient and easy to use. The doctor information was also rated positively, as most users found it clear and helpful.

For appointment management, most respondents gave a rating of 5 by showing that managing appointments is simple and effective. Overall, the results show that the system features provide a good user experience.

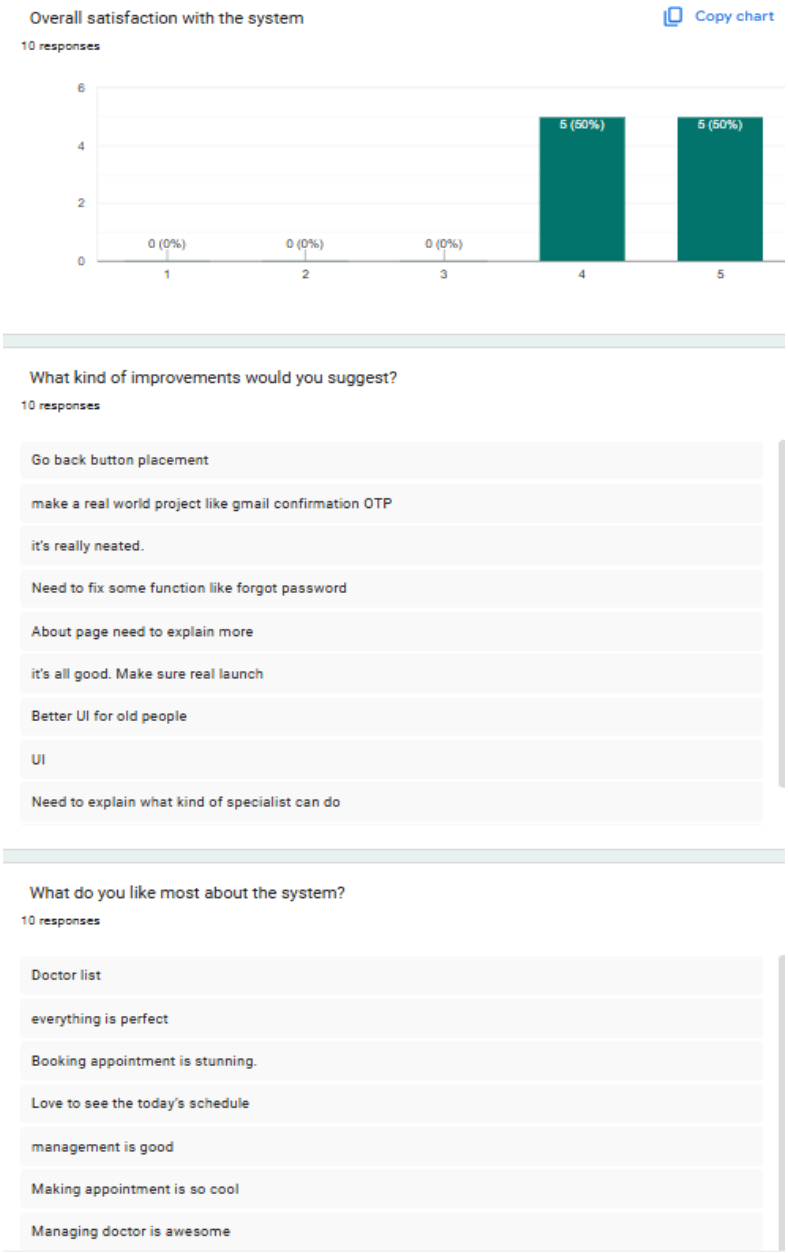


Figure 5.2.7 Overall Satisfaction and User Feedback

This figure shows users' overall satisfaction and feedback on the system. All respondents gave positive ratings with 50% giving a score of 4 and 50% giving a score of 5. This shows a high level of user satisfaction.

For improvements, users suggested making the back button placement better, adding features such as email or OTP verification, fixing the "Forgot Password" function and improving the user interface. Some users also recommended providing clearer explanations for doctor specializations and system pages.

On the positive side, users liked the doctor list, appointment booking process, and appointment management features. They described the system as easy to use and effective. Overall, the feedback shows that the system works well, with only a few minor improvements needed.

Unit Testing

Unit testing was performed during system development to ensure that each component works correctly. This process helps detect and resolve errors early by ensuring that all features of the Smart Clinic Booking System function as intended.

Table 5.2.1 Unit Testing Result

Feature	Tested Aspects	Status
Role-Based Login	Different access for admin and patient	Pass
User Registration	Register new users with validation	Pass
User Profile	Store and retrieve user information	Pass

Table 5.2.2 Admin Testing

Feature	Tested Aspects	Status
Doctor Management	Add, edit and delete doctor details	Pass
Appointment Management	Approve or reject appointments	Pass
Schedule Viewing	View schedule	Pass

Table 5.2.3 Patient Testing

Feature	Tested Aspects	Status
View Doctors	Display doctor list by specialization	Pass
Book Appointment	Select date and time for booking	Pass
Manage Appointment	View and cancel appointment	Pass

5.3 Achievement

The objectives of this project were successfully achieved through the development and evaluation of the Smart Clinic Booking System.

The first objective was completed through research and analysis of current systems. This helped identify important features, limitations and system requirements needed for effective appointment booking and management.

The second objective was achieved by creating the system structure using UML diagrams such as use case diagrams, activity diagrams and sequence diagrams. These designs show how different parts of the system work together to support appointment booking and management.

The third objective was achieved through system testing and user feedback. Participants completed tasks and gave feedback through a questionnaire. The results show that users found

the system easy to use, well-organized, and effective. Overall, the system provides a user-friendly experience for both patients and administrators.

In conclusion, the project successfully developed and evaluated a system that meets all the defined objectives and requirements.

5.4 Limitations

Despite the successful development of the Smart Clinic Booking System, there are some limitations. First, the system does not include advanced security features such as email verification or OTP authentication which may affect account security. Second, the “Forgot Password” function has some minor issues and needs improvement.

In addition, the system has limited scalability and is mainly designed for small clinics. The user interface may also not be fully suitable for all users, especially elderly users.

5.5 Future Work

In the future, several improvements can be made to make the system better and more useful. First, security can be improved by adding features such as email verification and OTP authentication to protect user accounts. The “Forgot Password” function should also be fixed and improved so that users can reset their passwords easily without any problems.

The user interface can be improved to make it more simple, clear, and suitable for all users, including elderly people. For example, larger buttons, clearer text and better navigation can help users understand the system more easily.

In addition, new features can be added to improve user experience. For example, a chatbot can be included to help users with common questions, guide them through the booking process and provide quick support. Notification reminders can also be added to remind patients about their upcoming appointments through email or SMS.

The system can also be improved by making it mobile-friendly so that users can easily access it from their smartphones. Real-time updates can be added to show available time slots instantly and avoid booking conflicts.

Furthermore, the system can be expanded to support larger clinics or hospitals with more doctors, departments and patients. More advanced features such as online payment, medical record storage and reporting systems can also be added to make the system more complete and useful.

Reference

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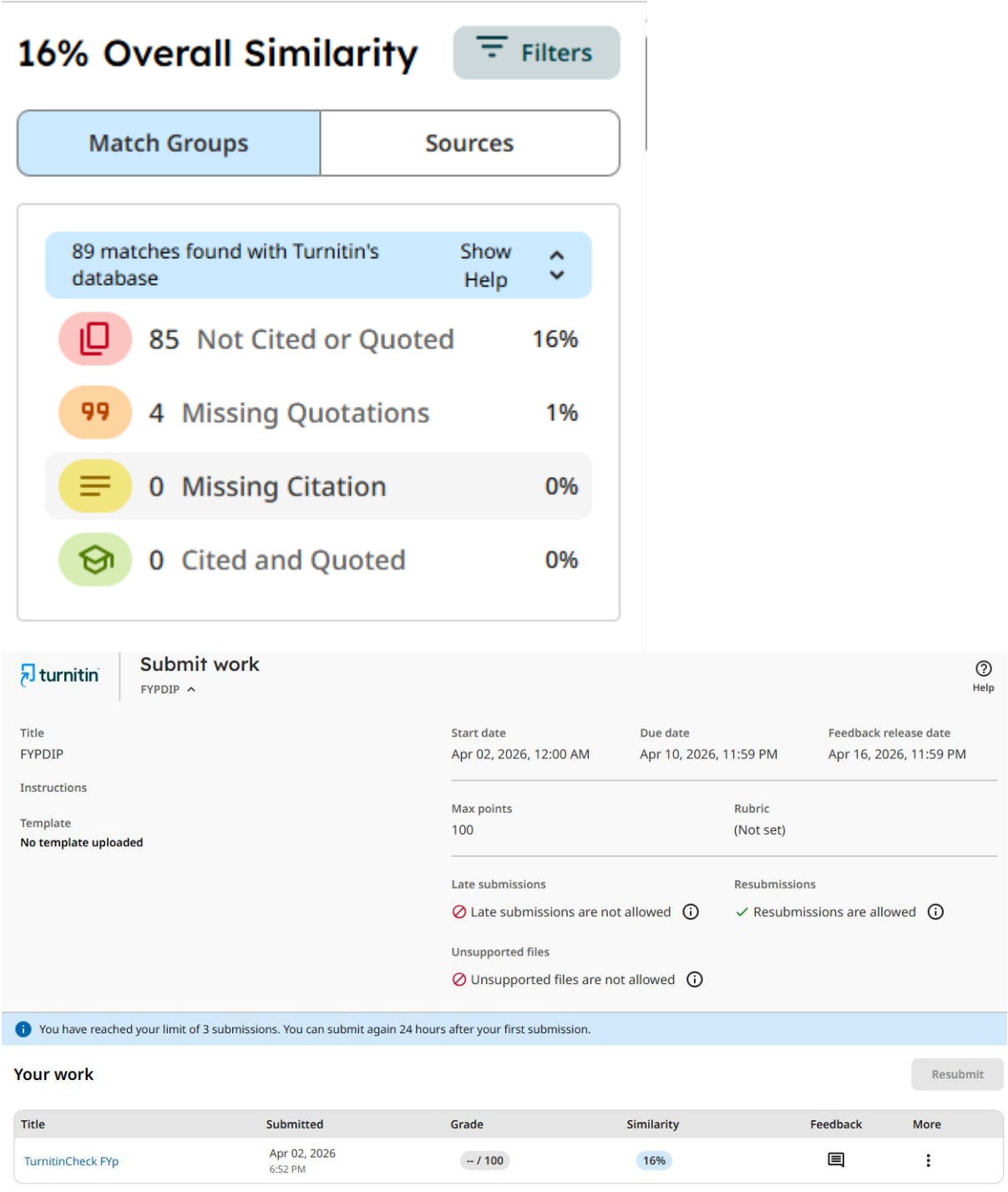
4. X. Zhang, T. Yu, and Y. Yan, “Design of hospital online appointment system,” International Journal of Healthcare Technology, vol. 5, no. 2, pp. 45–52, 2018.

5.I. Sommerville, Software Engineering, 10th ed. Boston, MA: Pearson, 2016.

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Appendices

Appendix 1: Turnitin



Appendix 2: Journals

International Journal of Research Publication and Reviews, Vol 6, Issue 3, pp 7618-7623 March 2025

WWW.IJRPR.COM

International Journal of Research Publication and Reviews

Journal homepage: www.ijrpr.com ISSN 2582-7421

Smart Healthcare Appointment System

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DOI : https://doi.org/10.55248/gengpi.6.0325.12149

ABSTRACT

The Doctor Appointment Booking System is a web-based platform designed to streamline healthcare scheduling. Built using the MERN Stack (MongoDB, Express.js, React, Node.js), it allows patients to book, reschedule, or cancel appointments seamlessly. Doctors manage their availability, and admins oversee operations. The system includes real-time scheduling, payment integration, automated notifications, and a blood donation module, ensuring an efficient and user-friendly healthcare experience. Additionally, the system enhances accessibility by providing a user-friendly dashboard for tracking appointments and managing doctor schedules. It reduces dependency on manual record-keeping, minimizes errors, and offers a secure platform for online transactions. The integration of cloud storage ensures that patient and doctor data remain accessible and protected at all times. The Doctor Appointment Booking System is a web-based platform designed to streamline healthcare scheduling. Built using the MERN Stack (MongoDB, Express.js, React, Node.js), it allows patients to book, reschedule, or cancel appointments seamlessly. Doctors manage their availability, and admins oversee operations. The system includes real-time scheduling, payment integration, automated notifications, and a blood donation module, ensuring an efficient and user-friendly healthcare experience.

Keywords: IoT, Plant Water Monitoring, Smart Irrigation, Soil Moisture Sensor, ESP8266, Automation.

1. Introduction

The Doctor Appointment Booking System is a comprehensive healthcare management platform developed to digitize and streamline appointment scheduling. Using the MERN Stack (MongoDB, Express.js, React, Node.js), the system ensures a seamless connection between patients, doctors, and administrators. Patients can easily browse available doctors, select suitable time slots, and book appointments in real-time, reducing the hassle of long waiting times and manual scheduling.

The system incorporates secure role-based authentication, allowing patients to manage their bookings, doctors to handle their schedules, and admins to oversee user and system operations. To further enhance accessibility, the system features automated notifications, online payment integration, and a blood donation module that connects donors with recipients in need. Additionally, the cloud-based architecture ensures high availability, scalability, and data security, making it an efficient and reliable solution for modern healthcare facilities.

The Doctor Appointment Booking System is a web-based application developed using the MERN Stack to facilitate seamless appointment scheduling between patients and doctors. The system provides an intuitive interface for patients to book appointments, doctors to manage their schedules, and administrators to oversee the entire platform. Additional features include secure authentication, real-time availability tracking, payment integration, and a blood donation module to help users find or donate blood when needed.

2. SYSTEM STUDY

2.1 EXISTING SYSTEM

INTERNATIONAL JOURNAL OF ACADEMIC RESEARCH IN BUSINESS & SOCIAL SCIENCES

Vol 14, Issue 2, (2024) E-ISSN: 2222-6990

An Implementation of Digital Platform to Enhance the Appointment Scheduling System

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To Link this Article: http://dx.doi.org/10.6007/IJARBSS/v14-i2/20635 DOI:10.6007/IJARBSS/v14-i2/20635

Published Date: 14 February 2024

Abstract

The high request for appointments and manually arranging the customer appointments became challenges for staff to manage the schedules. There was an increasing demand for customers visiting Company Z which lacked a proper platform to manage and keep the customer appointments. The purpose of the study is to implement a digital platform to enhance the appointment scheduling system. It was important to know and understand the practice and current process of appointment scheduling to improve the overall system to benefit the staff and customers. The action plan or intervention was proposed to use the digitalisation of the appointment scheduling system. The research design of the data collection method was qualitative data for semi-structured interviews and quantitative data for structured surveys. The intervention in cycle one did not achieve the targeted value due to the short period of intervention. Therefore, the intervention was improvised by categorising the services provided and simplifying the user interface to ease the process of scheduling appointments for cycle two. The overall interventions were successfully implemented and the new system enhanced the company service quality by reducing the customer wait time and streamlining the staff workflows.

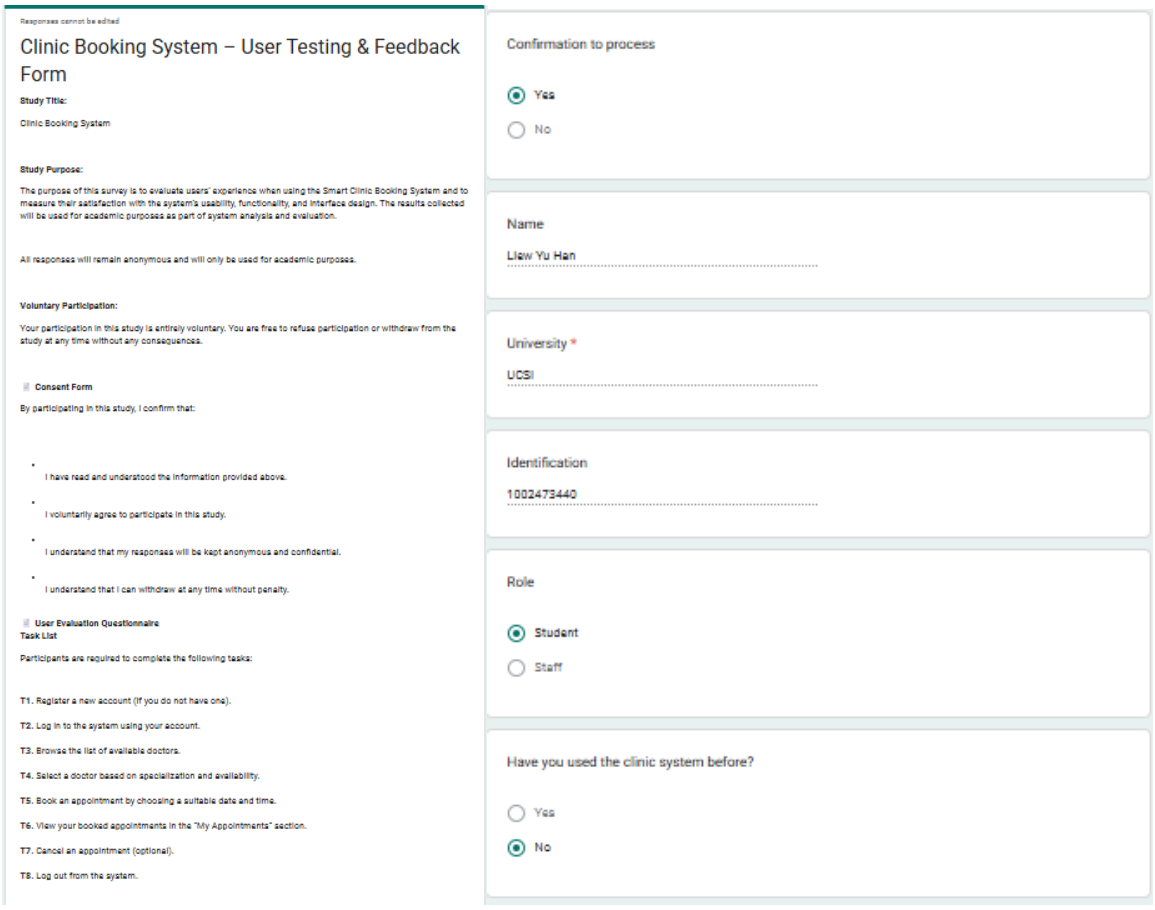
Keywords: Appointment Scheduling, Digitalisation, Action Research

Introduction

The service industry is a business sector in that part of the economy that provides services to customers rather than tangible objects. Every business not only sells goods but mostly serves services to people. A system is a group of components that work together to form a network

Appendix 3: Survey

Survey 1:



Responses cannot be edited

Clinic Booking System – User Testing & Feedback Form

Study Title:

Clinic Booking System

Study Purpose:

The purpose of this survey is to evaluate users' experience when using the Smart Clinic Booking System and to measure their satisfaction with the system's usability, functionality, and interface design. The results collected will be used for academic purposes as part of system analysis and evaluation.

All responses will remain anonymous and will only be used for academic purposes.

Voluntary Participation:

Your participation in this study is entirely voluntary. You are free to refuse participation or withdraw from the study at any time without any consequences.

Consent Form

By participating in this study, I confirm that:

- I have read and understood the information provided above.
- I voluntarily agree to participate in this study.
- I understand that my responses will be kept anonymous and confidential.
- I understand that I can withdraw at any time without penalty.

User Evaluation Questionnaire

Task List

Participants are required to complete the following tasks:

T1. Register a new account (if you do not have one).

T2. Log in to the system using your account.

T3. Browse the list of available doctors.

T4. Select a doctor based on specialization and availability.

T5. Book an appointment by choosing a suitable date and time.

T6. View your booked appointments in the "My Appointments" section.

T7. Cancel an appointment (optional).

T8. Log out from the system.

Confirmation to process

Yes

No

Name

Liew Yu Han

University *

UCSI

Identification

1002473440

Role

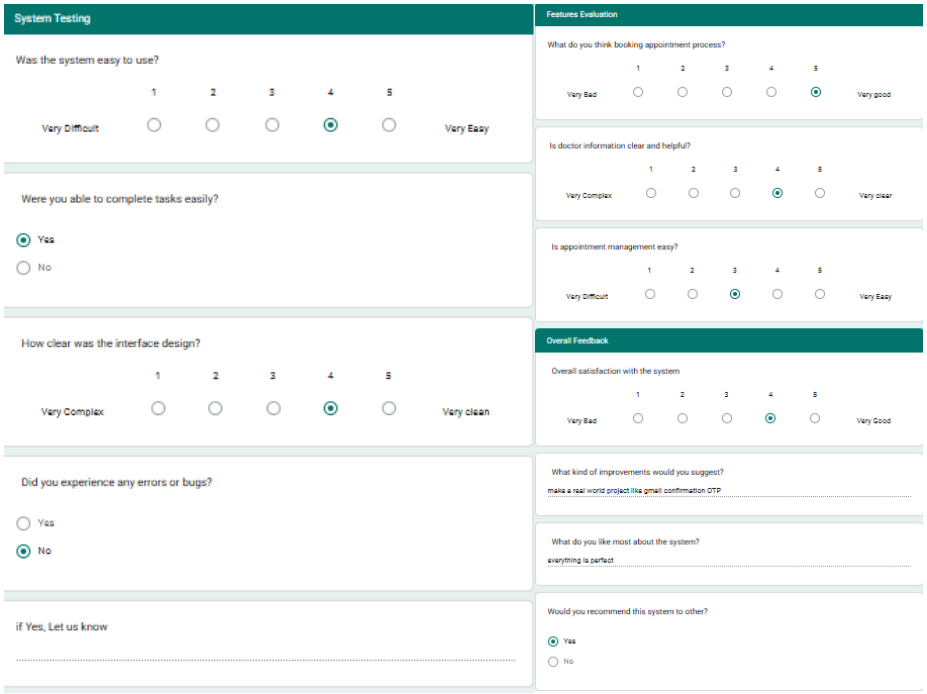
Student

Staff

Have you used the clinic system before?

Yes

No



System Testing

Was the system easy to use?

1 2 3 4 5

Very Difficult — Very Easy

Were you able to complete tasks easily?

Yes

No

How clear was the interface design?

1 2 3 4 5

Very Complex — Very clean

Did you experience any errors or bugs?

Yes

No

if Yes, Let us know

Features Evaluation

What do you think booking appointment process?

1 2 3 4 5

Very Bad — Very good

Is doctor information clear and helpful?

1 2 3 4 5

Very Complex — Very clear

Is appointment management easy?

1 2 3 4 5

Very Difficult — Very Easy

Overall Feedback

Overall satisfaction with the system

1 2 3 4 5

Very Bad — Very Good

What kind of improvements would you suggest?

make a real world project like gmail confirmation OTP

What do you like most about the system?

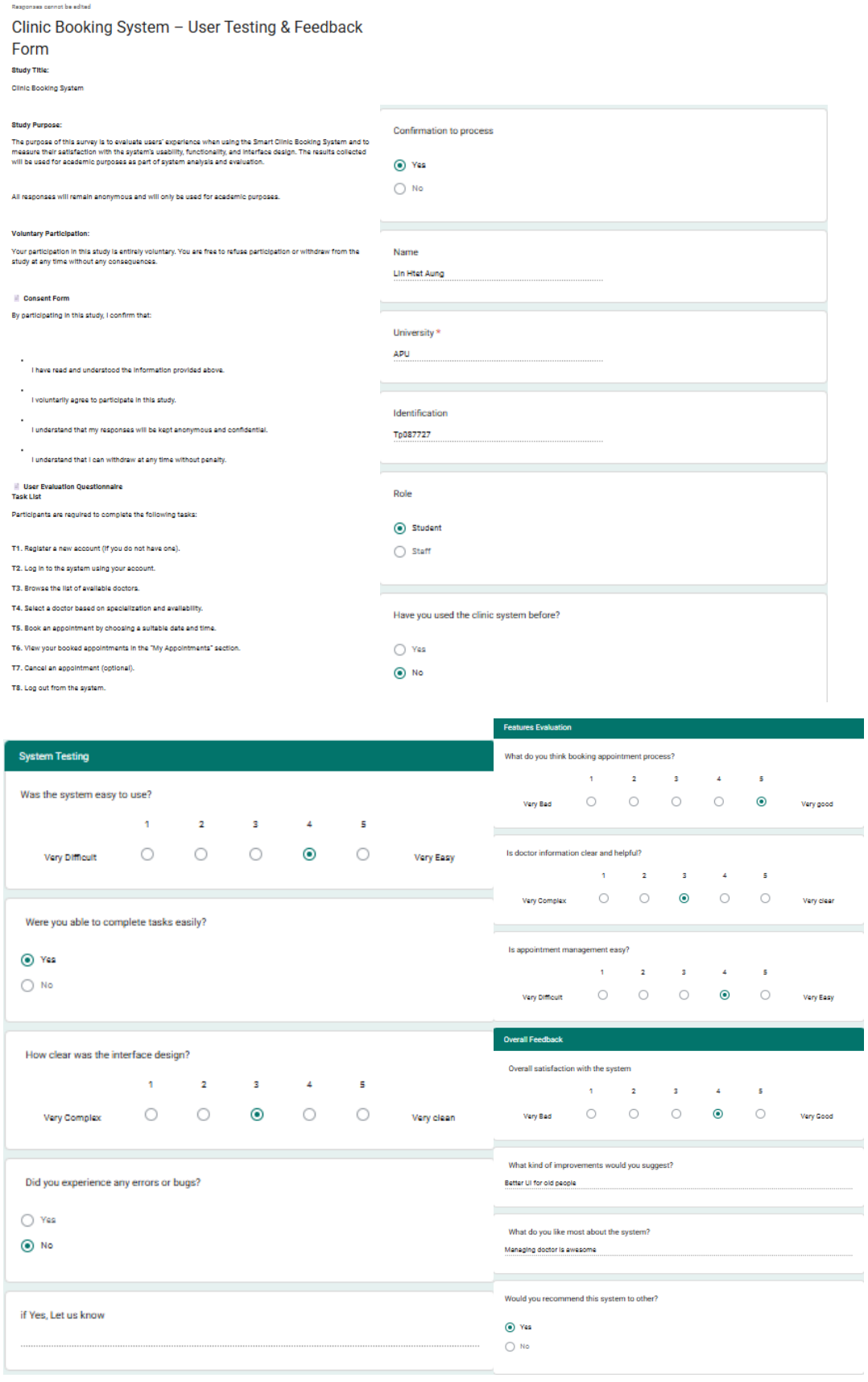
everything is perfect

Would you recommend this system to other?

Yes

No

Survey 2:



Responses cannot be edited

Clinic Booking System – User Testing & Feedback Form

Study Title:

Clinic Booking System

Study Purpose:

The purpose of this survey is to evaluate users' experience when using the Smart Clinic Booking System and to measure their satisfaction with the system's usability, functionality, and interface design. The results collected will be used for academic purposes as part of system analysis and evaluation.

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User Evaluation Questionnaire

Task List

Participants are required to complete the following tasks:

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T4. Select a doctor based on specialization and availability.

T5. Book an appointment by choosing a suitable date and time.

T6. View your booked appointments in the "My Appointments" section.

T7. Cancel an appointment (optional).

T8. Log out from the system.

Confirmation to process

Yes

No

Name

Lin Htet Aung

University *

APU

Identification

Tp087727

Role

Student

Staff

Have you used the clinic system before?

Yes

No

System Testing

Was the system easy to use?

	1	2	3	4	5	
Very Difficult	○	○	○	◉	○	Very Easy

Were you able to complete tasks easily?

Yes

No

How clear was the interface design?

	1	2	3	4	5	
Very Complex	○	○	◉	○	○	Very clean

Did you experience any errors or bugs?

Yes

No

if Yes, Let us know

Features Evaluation

What do you think booking appointment process?

	1	2	3	4	5	
Very Bad	○	○	○	○	◉	Very good

Is doctor information clear and helpful?

	1	2	3	4	5	
Very Complex	○	○	◉	○	○	Very clear

Is appointment management easy?

	1	2	3	4	5	
Very Difficult	○	○	○	◉	○	Very Easy

Overall Feedback

Overall satisfaction with the system

	1	2	3	4	5	
Very Bad	○	○	○	◉	○	Very Good

What kind of improvements would you suggest?

Better UI for old people

What do you like most about the system?

Managing doctor is awesome

Would you recommend this system to other?

Yes

No

Survey 3:

Responses cannot be edited

Clinic Booking System – User Testing & Feedback Form

Study Title:

Clinic Booking System

Study Purpose:

The purpose of this survey is to evaluate users' experience when using the Smart Clinic Booking System and to measure their satisfaction with the system's usability, functionality, and interface design. The results collected will be used for academic purposes as part of system analysis and evaluation.

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Voluntary Participation:

Your participation in this study is entirely voluntary. You are free to refuse participation or withdraw from the study at any time without any consequences.

Consent Form

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- I understand that my responses will be kept anonymous and confidential.
- I understand that I can withdraw at any time without penalty.

User Evaluation Questionnaire

Task List

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T3. Browse the list of available doctors.

T4. Select a doctor based on specialization and availability.

T5. Book an appointment by choosing a suitable date and time.

T6. View your booked appointments in the "My Appointments" section.

T7. Cancel an appointment (optional).

T8. Log out from the system.

Confirmation to process

- (•) Yes
- ( ) No

Name

Pyae Phyo Han

University *

APU

Identification

Tp086863

Role

- (•) Student
- ( ) Staff

Have you used the clinic system before?

- (•) Yes
- ( ) No

System Testing

Was the system easy to use?

	1	2	3	4	5	
Very Difficult	○	○	○	○	◉	Very Easy

Were you able to complete tasks easily?

- (•) Yes
- ( ) No

How clear was the interface design?

	1	2	3	4	5	
Very Complex	○	○	◉	○	○	Very clean

Did you experience any errors or bugs?

- (•) Yes
- ( ) No

if Yes, Let us know

Forgot password is not functioning.

Features Evaluation

What do you think booking appointment process?

	1	2	3	4	5	
Very Bad	○	○	○	◉	○	Very good

Is doctor information clear and helpful?

	1	2	3	4	5	
Very Complex	○	○	○	○	◉	Very clear

Is appointment management easy?

	1	2	3	4	5	
Very Difficult	○	○	○	○	◉	Very Easy

Overall Feedback

Overall satisfaction with the system

	1	2	3	4	5	
Very Bad	○	○	○	○	◉	Very Good

What kind of improvements would you suggest?

Need to fix some function like forgot password

What do you like most about the system?

Love to see the today's schedule

Would you recommend this system to other?

- (•) Yes
- ( ) No

Survey 4:

Responses cannot be edited

Clinic Booking System – User Testing & Feedback Form

Study Title:

Clinic Booking System

Study Purpose:

The purpose of this survey is to evaluate users' experience when using the Smart Clinic Booking System and to measure their satisfaction with the system's usability, functionality, and interface design. The results collected will be used for academic purposes as part of system analysis and evaluation.

All responses will remain anonymous and will only be used for academic purposes.

Voluntary Participation:

Your participation in this study is entirely voluntary. You are free to refuse participation or withdraw from the study at any time without any consequences.

Consent Form

By participating in this study, I confirm that:

- I have read and understood the information provided above.
- I voluntarily agree to participate in this study.
- I understand that my responses will be kept anonymous and confidential.
- I understand that I can withdraw at any time without penalty.

User Evaluation Questionnaire

Task List

Participants are required to complete the following tasks:

T1. Register a new account (if you do not have one).

T2. Log in to the system using your account.

T3. Browse the list of available doctors.

T4. Select a doctor based on specialization and availability.

T5. Book an appointment by choosing a suitable date and time.

T6. View your booked appointments in the "My Appointments" section.

T7. Cancel an appointment (optional).

T8. Log out from the system.

Confirmation to process

- (•) Yes
- ( ) No

Name

Aung Kaung Khant

University *

APU

Identification

Tp088740

Role

- (•) Student
- ( ) Staff

Have you used the clinic system before?

- ( ) Yes
- (•) No

System Testing

Was the system easy to use?

	1	2	3	4	5	
Very Difficult	○	○	○	◉	○	Very Easy

Were you able to complete tasks easily?

- (•) Yes
- ( ) No

How clear was the interface design?

	1	2	3	4	5	
Very Complex	○	○	○	◉	○	Very clean

Did you experience any errors or bugs?

- ( ) Yes
- (•) No

if Yes, Let us know

Features Evaluation

What do you think booking appointment process?

	1	2	3	4	5	
Very Bad	○	○	○	◉	○	Very good

Is doctor information clear and helpful?

	1	2	3	4	5	
Very Complex	○	○	○	◉	○	Very clear

Is appointment management easy?

	1	2	3	4	5	
Very Difficult	○	○	◉	○	○	Very Easy

Overall Feedback

Overall satisfaction with the system

	1	2	3	4	5	
Very Bad	○	○	○	◉	○	Very Good

What kind of improvements would you suggest?

UI

What do you like most about the system?

Booking process

Would you recommend this system to other?

- (•) Yes
- ( ) No

Survey 5:

Responses cannot be edited

Clinic Booking System – User Testing & Feedback Form

Study Title:

Clinic Booking System

Study Purpose:

The purpose of this survey is to evaluate users' experience when using the Smart Clinic Booking System and to measure their satisfaction with the system's usability, functionality, and interface design. The results collected will be used for academic purposes as part of system analysis and evaluation.

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- I understand that I can withdraw at any time without penalty.

User Evaluation Questionnaire

Task List

Participants are required to complete the following tasks:

T1. Register a new account (if you do not have one).

T2. Log in to the system using your account.

T3. Browse the list of available doctors.

T4. Select a doctor based on specialization and availability.

T5. Book an appointment by choosing a suitable date and time.

T6. View your booked appointments in the "My Appointments" section.

T7. Cancel an appointment (optional).

T8. Log out from the system.

Confirmation to process

- (•) Yes
- ( ) No

Name

Shin Thant Aung

University *

APU

Identification

Tp089833

Role

- (•) Student
- ( ) Staff

Have you used the clinic system before?

- (•) Yes
- ( ) No

System Testing

Was the system easy to use?

	1	2	3	4	5	
Very Difficult	○	○	○	◉	○	Very Easy

Were you able to complete tasks easily?

- (•) Yes
- ( ) No

How clear was the interface design?

	1	2	3	4	5	
Very Complex	○	○	◉	○	○	Very clean

Did you experience any errors or bugs?

- ( ) Yes
- (•) No

if Yes, Let us know

.....

Features Evaluation

What do you think booking appointment process?

	1	2	3	4	5	
Very Bad	○	○	○	○	◉	Very good

Is doctor information clear and helpful?

	1	2	3	4	5	
Very Complex	○	○	○	○	◉	Very clear

Is appointment management easy?

	1	2	3	4	5	
Very Difficult	○	○	○	◉	○	Very Easy

Overall Feedback

Overall satisfaction with the system

	1	2	3	4	5	
Very Bad	○	○	○	○	◉	Very Good

What kind of improvements would you suggest?

button click

What do you like most about the system?

Booking management

Would you recommend this system to other?

- (•) Yes
- ( ) No

Survey 6:

Responses cannot be edited

Clinic Booking System – User Testing & Feedback Form

Study Title:

Clinic Booking System

Study Purpose:

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Consent Form

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- I voluntarily agree to participate in this study.
- I understand that my responses will be kept anonymous and confidential.
- I understand that I can withdraw at any time without penalty.

User Evaluation Questionnaire

Task List

Participants are required to complete the following tasks:

T1. Register a new account (if you do not have one).

T2. Log in to the system using your account.

T3. Browse the list of available doctors.

T4. Select a doctor based on specialization and availability.

T5. Book an appointment by choosing a suitable date and time.

T6. View your booked appointments in the "My Appointments" section.

T7. Cancel an appointment (optional).

T8. Log out from the system.

Confirmation to process

- (•) Yes
- ( ) No

Name

Thea Khai Jian

University *

UCSI

Identification

1002473256

Role

- (•) Student
- ( ) Staff

Have you used the clinic system before?

- (•) Yes
- ( ) No

System Testing

Was the system easy to use?

	1	2	3	4	5	
Very Difficult	○	○	○	◉	○	Very Easy

Were you able to complete tasks easily?

- (•) Yes
- ( ) No

How clear was the interface design?

	1	2	3	4	5	
Very Complex	○	○	○	◉	○	Very clean

Did you experience any errors or bugs?

- ( ) Yes
- (•) No

if Yes, Let us know

.....

Features Evaluation

What do you think booking appointment process?

	1	2	3	4	5	
Very Bad	○	○	◉	○	○	Very good

Is doctor information clear and helpful?

	1	2	3	4	5	
Very Complex	○	○	○	○	◉	Very clear

Is appointment management easy?

	1	2	3	4	5	
Very Difficult	○	○	○	○	◉	Very Easy

Overall Feedback

Overall satisfaction with the system

	1	2	3	4	5	
Very Bad	○	○	○	◉	○	Very Good

What kind of improvements would you suggest?

Go back button placement

What do you like most about the system?

Doctor list

Would you recommend this system to other?

- (•) Yes
- ( ) No

Survey 7:

Responses cannot be edited

Clinic Booking System – User Testing & Feedback Form

Study Title:

Clinic Booking System

Study Purpose:

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- I understand that I can withdraw at any time without penalty.

User Evaluation Questionnaire

Task List

Participants are required to complete the following tasks:

T1. Register a new account (if you do not have one).

T2. Log in to the system using your account.

T3. Browse the list of available doctors.

T4. Select a doctor based on specialization and availability.

T5. Book an appointment by choosing a suitable date and time.

T6. View your booked appointments in the "My Appointments" section.

T7. Cancel an appointment (optional).

T8. Log out from the system.

Confirmation to process

- (•) Yes
- ( ) No

Name

Kaung Khant Thaw

University *

Taylor University

Identification

0361581

Role

- (•) Student
- ( ) Staff

Have you used the clinic system before?

- (•) Yes
- ( ) No

System Testing

Was the system easy to use?

	1	2	3	4	5	
Very Difficult	○	○	○	◉	○	Very Easy

Were you able to complete tasks easily?

- (•) Yes
- ( ) No

How clear was the interface design?

	1	2	3	4	5	
Very Complex	○	○	○	◉	○	Very clean

Did you experience any errors or bugs?

- ( ) Yes
- (•) No

if Yes, Let us know

.....

Features Evaluation

What do you think booking appointment process?

	1	2	3	4	5	
Very Bad	○	○	○	○	◉	Very good

Is doctor information clear and helpful?

	1	2	3	4	5	
Very Complex	○	○	○	◉	○	Very clear

Is appointment management easy?

	1	2	3	4	5	
Very Difficult	○	○	○	○	◉	Very Easy

Overall Feedback

Overall satisfaction with the system

	1	2	3	4	5	
Very Bad	○	○	○	○	◉	Very Good

What kind of improvements would you suggest?

It's all good. Make sure real launch

What do you like most about the system?

Making appointment is so cool

Would you recommend this system to other?

- (•) Yes
- ( ) No

Survey 8:

Responses cannot be edited

Clinic Booking System – User Testing & Feedback Form

Study Title:

Clinic Booking System

Study Purpose:

The purpose of this survey is to evaluate users' experience when using the Smart Clinic Booking System and to measure their satisfaction with the system's usability, functionality, and interface design. The results collected will be used for academic purposes as part of system analysis and evaluation.

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User Evaluation Questionnaire

Task List

Participants are required to complete the following tasks:

T1. Register a new account (if you do not have one).

T2. Log in to the system using your account.

T3. Browse the list of available doctors.

T4. Select a doctor based on specialization and availability.

T5. Book an appointment by choosing a suitable date and time.

T6. View your booked appointments in the "My Appointments" section.

T7. Cancel an appointment (optional).

T8. Log out from the system.

Confirmation to process

- (•) Yes
- ( ) No

Name

Muhammad Haikal Azmi

University *

UCSI

Identification

1002372636

Role

- (•) Student
- ( ) Staff

Have you used the clinic system before?

- (•) Yes
- ( ) No

System Testing

Was the system easy to use?

	1	2	3	4	5	
Very Difficult	○	○	○	◉	○	Very Easy

Were you able to complete tasks easily?

- (•) Yes
- ( ) No

How clear was the interface design?

	1	2	3	4	5	
Very Complex	○	○	○	◉	○	Very clean

Did you experience any errors or bugs?

- ( ) Yes
- (•) No

if Yes, Let us know

Features Evaluation

What do you think booking appointment process?

	1	2	3	4	5	
Very Bad	○	○	○	◉	○	Very good

Is doctor information clear and helpful?

	1	2	3	4	5	
Very Complex	○	○	○	○	◉	Very clear

Is appointment management easy?

	1	2	3	4	5	
Very Difficult	○	○	○	○	◉	Very Easy

Overall Feedback

Overall satisfaction with the system

	1	2	3	4	5	
Very Bad	○	○	○	○	◉	Very Good

What kind of improvements would you suggest?

It's really neated.

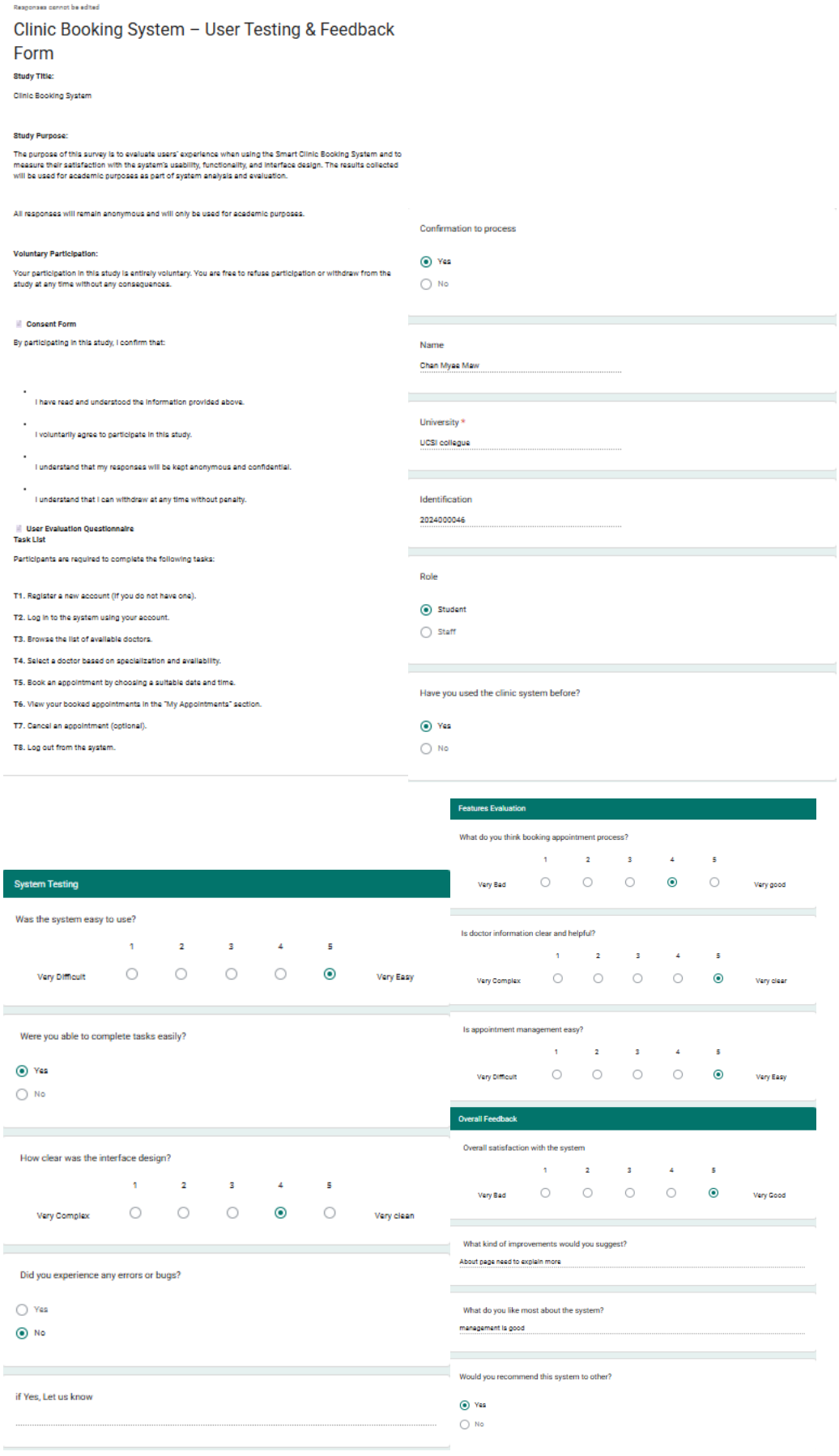
What do you like most about the system?

Booking appointment is stunning.

Would you recommend this system to other?

- (•) Yes
- ( ) No

Survey 9:



Responses cannot be edited

Clinic Booking System – User Testing & Feedback Form

Study Title:

Clinic Booking System

Study Purpose:

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User Evaluation Questionnaire

Task List

Participants are required to complete the following tasks:

T1. Register a new account (if you do not have one).

T2. Log in to the system using your account.

T3. Browse the list of available doctors.

T4. Select a doctor based on specialization and availability.

T5. Book an appointment by choosing a suitable date and time.

T6. View your booked appointments in the "My Appointments" section.

T7. Cancel an appointment (optional).

T8. Log out from the system.

Confirmation to process

- (•) Yes
- ( ) No

Name

Chan Myae Maw

University *

UCSI collegue

Identification

2024000046

Role

- (•) Student
- ( ) Staff

Have you used the clinic system before?

- (•) Yes
- ( ) No

System Testing

Was the system easy to use?

	1	2	3	4	5	
Very Difficult	○	○	○	○	(•)	Very Easy

Were you able to complete tasks easily?

- (•) Yes
- ( ) No

How clear was the interface design?

	1	2	3	4	5	
Very Complex	○	○	○	(•)	○	Very clean

Did you experience any errors or bugs?

- ( ) Yes
- (•) No

if Yes, Let us know

Features Evaluation

What do you think booking appointment process?

	1	2	3	4	5	
Very Bad	○	○	○	(•)	○	Very good

Is doctor information clear and helpful?

	1	2	3	4	5	
Very Complex	○	○	○	○	(•)	Very clear

Is appointment management easy?

	1	2	3	4	5	
Very Difficult	○	○	○	○	(•)	Very Easy

Overall Feedback

Overall satisfaction with the system

	1	2	3	4	5	
Very Bad	○	○	○	○	(•)	Very Good

What kind of improvements would you suggest?

About page need to explain more

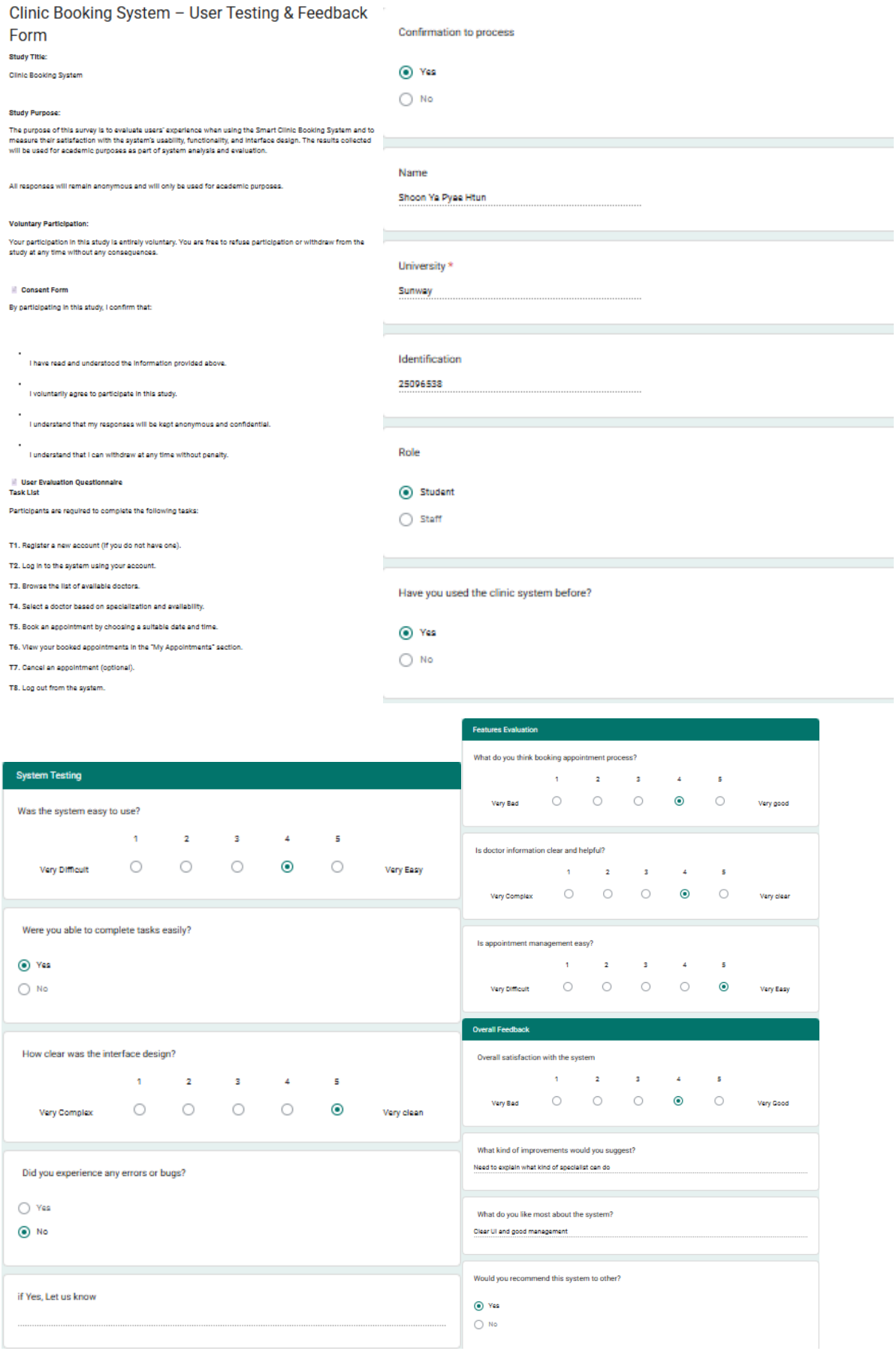
What do you like most about the system?

management is good

Would you recommend this system to other?

- (•) Yes
- ( ) No

Survey 10:



Clinic Booking System – User Testing & Feedback Form

Study Title:

Clinic Booking System

Study Purpose:

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- I understand that my responses will be kept anonymous and confidential.
- I understand that I can withdraw at any time without penalty.

User Evaluation Questionnaire

Task List

Participants are required to complete the following tasks:

T1. Register a new account (if you do not have one).

T2. Log in to the system using your account.

T3. Browse the list of available doctors.

T4. Select a doctor based on specialization and availability.

T5. Book an appointment by choosing a suitable date and time.

T6. View your booked appointments in the "My Appointments" section.

T7. Cancel an appointment (optional).

T8. Log out from the system.

Confirmation to process

- (•) Yes
- ( ) No

Name

Shoon Ya Pyae Htun

University *

Sunway

Identification

25096538

Role

- (•) Student
- ( ) Staff

Have you used the clinic system before?

- (•) Yes
- ( ) No

System Testing

Was the system easy to use?

	1	2	3	4	5	
Very Difficult	○	○	○	◉	○	Very Easy

Were you able to complete tasks easily?

- (•) Yes
- ( ) No

How clear was the interface design?

	1	2	3	4	5	
Very Complex	○	○	○	○	◉	Very clean

Did you experience any errors or bugs?

- ( ) Yes
- (•) No

if Yes, Let us know

Features Evaluation

What do you think booking appointment process?

	1	2	3	4	5	
Very Bad	○	○	○	◉	○	Very good

Is doctor information clear and helpful?

	1	2	3	4	5	
Very Complex	○	○	○	◉	○	Very clear

Is appointment management easy?

	1	2	3	4	5	
Very Difficult	○	○	○	○	◉	Very Easy

Overall Feedback

Overall satisfaction with the system

	1	2	3	4	5	
Very Bad	○	○	○	◉	○	Very Good

What kind of improvements would you suggest?

Need to explain what kind of specialist can do

What do you like most about the system?

Clear UI and good management

Would you recommend this system to other?

- (•) Yes
- ( ) No

Code

index.php

```php
<?php include 'menu.php'; ?>

<style>
.hero {
    background: linear-gradient(to right, #0d6efd, #4dabf7);
    color: white;
    padding: 80px 20px;
    border-radius: 12px;
}
.feature-card {
    transition: 0.3s;
    border-radius: 12px;
}
.feature-card:hover {
    transform: translateY(-5px);
}
.icon {
    font-size: 40px;
}
</style>

<div class="container mt-4">
    <div class="hero text-center">
        <h1 class="display-5 fw-bold">🏥 Smart Clinic Booking System</h1>
        <p class="lead mt-3">
            A modern healthcare solution designed to simplify appointment scheduling
            and improve patient experience through efficient digital services.
        </p>
        <a href="login_select.php" class="btn btn-light btn-lg m-2">Login</a>
        <a href="register.php" class="btn btn-success btn-lg m-2">Register</a>
        <p class="mt-3">
            Providing convenient access to healthcare services anytime, anywhere.
        </p>
    </div>
    <div class="row text-center mt-5 g-4">
        <div class="col-md-4">
            <div class="card p-4 shadow feature-card">
                <div class="icon">👨‍⚕️</div>
                <h4 class="mt-3">Experienced Doctors</h4>
                <p>
                    Our clinic is supported by qualified medical professionals dedicated to
                    delivering high-quality and reliable healthcare services.
                </p>
            </div>
        </div>
```

```php
        <div class="col-md-4">
            <div class="card p-4 shadow feature-card">
                <div class="icon">📅</div>
                <h4 class="mt-3">Online Appointment Booking</h4>
                <p>
                    Easily schedule appointments at your convenience with our user-friendly
                    and efficient online booking system.
                </p>
            </div>
        </div>

        <div class="col-md-4">
            <div class="card p-4 shadow feature-card">
                <div class="icon">⚡</div>
                <h4 class="mt-3">Efficient & Fast Service</h4>
                <p>
                    Minimize waiting time and receive prompt medical attention through our
                    streamlined appointment management process.
                </p>
            </div>
        </div>
    </div>
    <div class="text-center mt-5">
        <h4>Why Choose Us?</h4>
        <p class="text-muted">
            We combine technology and healthcare to provide a seamless experience
            for both patients and administrators, ensuring efficiency, accuracy,
            and convenience in every interaction.
        </p>
    </div>
    <footer class="text-center mt-5 p-3 bg-light rounded">
        <p class="mb-0">© 2026 Smart Clinic Booking System | All Rights Reserved</p>
    </footer>
</div>
```

about.php

```php
<?php include 'menu.php'; ?>  // navigation menu for users

<style>
//Main banner
.about-hero {
    background: linear-gradient(to right, #0d6efd, #4dabf7);
    color: white;
    padding: 60px 20px;
    border-radius: 12px;
    text-align: center;
}
//designing card
.section-card {
    border-radius: 12px;
    transition: 0.3s;
}
//slight effect
.section-card:hover {
    transform: translateY(-5px);
}
.icon {
    font-size: 35px;
}
</style>

<div class="container mt-4">
    <div class="about-hero">
        <h2>About Our Clinic</h2>
        <p class="mt-3">
            We are committed to providing high-quality healthcare services with modern
            technology to ensure convenience, efficiency, and patient satisfaction.
        </p>
    </div>
    <div class="row mt-5 g-4">
        <div class="col-md-4">
            <div class="card p-4 shadow section-card text-center">
                <div class="icon">🎯</div>
                <h4 class="mt-3">Our Mission</h4>
                <p>
                    To deliver accessible, reliable, and efficient healthcare services
                    through a user-friendly online system.
                </p>
            </div>
        </div>
        <div class="col-md-4">
            <div class="card p-4 shadow section-card text-center">
                <div class="icon">👁️</div>
                <h4 class="mt-3">Our Vision</h4>
                <p>
                    To become a leading smart healthcare provider by integrating
                    technology with medical services.
                </p>
            </div>
        </div>
        <div class="col-md-4">
            <div class="card p-4 shadow section-card text-center">
                <div class="icon">❤️</div>
                <h4 class="mt-3">Our Values</h4>
                <p>
                    We prioritize patient care, professionalism, integrity, and
                    continuous improvement in all our services.
                </p>
            </div>
        </div>
    </div>
    <div class="text-center mt-5">
        <h4>Why Choose Our Clinic?</h4>
        <p class="text-muted">
            Our clinic provides professional healthcare services with an efficient
            online appointment booking system to reduce waiting time and improve
            patient experience. We aim to make healthcare more accessible and convenient
            for everyone.
        </p>
    </div>
    <footer class="text-center mt-5 p-3 bg-light rounded">
        <p class="mb-0">© 2026 Smart Clinic Booking System</p>
    </footer>

</div>
```

contact.php

```php
<?php include 'menu.php'; ?>

<style>
.contact-hero {
    background: linear-gradient(to right, #0d6efd, #4dabf7);
    color: white;
    padding: 60px 20px;
    border-radius: 12px;
    text-align: center;
}
.contact-card {
    border-radius: 12px;
    transition: 0.3s;
}
.contact-card:hover {
    transform: translateY(-5px);
}
.icon {
    font-size: 30px;
}
</style>

<div class="container mt-4">

    <div class="contact-hero">
        <h2>Contact Us</h2>
        <p class="mt-3">
            Get in touch with us for any inquiries, support, or appointment assistance.
        </p>
    </div>
    <div class="row mt-5 g-4 text-center">
        <div class="col-md-4">
            <div class="card p-4 shadow contact-card">
                <div class="icon">📧</div>
                <h5 class="mt-3">Email</h5>
                <p>clinic@gmail.com</p>
            </div>
        </div>
        <div class="col-md-4">
            <div class="card p-4 shadow contact-card">
                <div class="icon">📞</div>
                <h5 class="mt-3">Phone</h5>
                <p>+60 123 456 789</p>
            </div>
        </div>
```

```php
        <div class="col-md-4">
            <div class="card p-4 shadow contact-card">
                <div class="icon">📍</div>
                <h5 class="mt-3">Address</h5>
                <p>Kuala Lumpur, Malaysia</p>
            </div>
        </div>
    </div>
    // contact form
    <div class="row mt-5">
        <div class="col-md-6 mx-auto">
            <div class="card p-4 shadow">
                <h4 class="text-center mb-3">Send Us a Message</h4>

                <form>
                    <div class="mb-3">
                        <input type="text" class="form-control" placeholder="Your Name">
                    </div>

                    <div class="mb-3">
                        <input type="email" class="form-control" placeholder="Your Email">
                    </div>

                    <div class="mb-3">
                        <textarea class="form-control" rows="4" placeholder="Your Message"></textarea>
                    </div>

                    <button class="btn btn-primary w-100">Send Message</button>
                </form>
            </div>
        </div>
    </div>
    <footer class="text-center mt-5 p-3 bg-light rounded">
        <p class="mb-0">© 2026 Smart Clinic Booking System</p>
    </footer>
</div>
```

menu.php

```php
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
<nav class="navbar navbar-expand-lg navbar-dark bg-primary">
    <div class="container-fluid">
        <a class="navbar-brand" href="index.php">Clinic System</a>
        <div>
            <a class="btn btn-light me-2" href="index.php">Home</a>
            <a class="btn btn-light me-2" href="about.php">About</a>
            <a class="btn btn-light" href="contact.php">Contact Us</a>
        </div>
    </div>
</nav>
<br>
```

register.php

```php
<?php
session_start();
include 'menu.php';
include 'db_connect.php';

if (isset($_POST['register'])) {

    $name = $_POST['name'];
    $email = $_POST['email'];
    $raw_password = $_POST['password'];
    $confirm_password = $_POST['confirm_password'];
    $phone = $_POST['phone'];
    if ($raw_password !== $confirm_password) // password check and confirmed one
    {
        echo "<div class='container mt-3'><div class='alert alert-danger'>
                Passwords do not match!
              </div></div>";
    }
    elseif (!preg_match("/^[a-zA-Z0-9._%+-]+@gmail\.com$/", $email))  // gmail check
    {
        echo "<div class='container mt-3'><div class='alert alert-danger'>
                Email must be a valid Gmail address (example@gmail.com).
              </div></div>";
    }
    elseif (!preg_match("/^\+?[0-9]+$/", $phone)) // phone check
    {
        echo "<div class='container mt-3'><div class='alert alert-danger'>
                Phone number can only contain numbers and optional '+' at the beginning.
              </div></div>";
    }
    elseif (!preg_match("/^(?=.*[A-Z]).{6,}$/", $raw_password)) // password min length check
    {
        echo "<div class='container mt-3'><div class='alert alert-danger'>
                Password must be at least 6 characters and contain at least one uppercase le
              </div></div>";
    }
    else  //hashing password for security
    {
        $password = password_hash($raw_password, PASSWORD_DEFAULT);

        $check = mysqli_query($conn, "SELECT * FROM users WHERE email='$email'");

        if (mysqli_num_rows($check) > 0) {

            echo "<div class='container mt-3'><div class='alert alert-warning'>
                    Email already registered!
                  </div></div>";

        } else {

            $sql = "INSERT INTO users (name, email, password, phone, role)
                    VALUES ('$name', '$email', '$password', '$phone', 'patient')";

            if (mysqli_query($conn, $sql)) {

                echo "<div class='container mt-3'><div class='alert alert-success'>
                        Registration Successful!
                      </div></div>";

            } else {

                echo "<div class='container mt-3'><div class='alert alert-danger'>
                        Database Error!
                      </div></div>";
            }
        }
    }
}
```

```php
<div class="container mt-5">
    <div class="row justify-content-center">
        <div class="col-md-6">
            <button onclick="history.back()" class="btn btn-outline-secondary mb-3">
                ← Back
            </button>

            <div class="card shadow p-4">

                <h3 class="text-center mb-4">📝 Client Registration</h3>

                <form method="POST">
                    <div class="mb-3">
                        <label class="form-label">Full Name</label>
                        <input type="text" name="name" class="form-control" required>
                    </div>
                    <div class="mb-3">
                        <label class="form-label">Email</label>
                        <input type="email" name="email" class="form-control" required>
                    </div>
                    <div class="mb-3">
                        <label class="form-label">Password</label>

                        <div class="input-group">
                            <input type="password" name="password" id="password" class="form-control" required>

                            <button type="button" class="btn btn-outline-secondary" onclick="togglePassword('password')">
                                👁
                            </button>
                        </div>

                        <small class="text-muted">
                            Must be at least 6 characters and include 1 uppercase letter
                        </small>
                    </div>
                    <div class="mb-3">  // confirm password
                        <label class="form-label">Confirm Password</label>

                        <div class="input-group">
                            <input type="password" name="confirm_password" id="confirm_password" class="form-control" required>

                            <button type="button" class="btn btn-outline-secondary" onclick="togglePassword('confirm_password')">
                                👁
                            </button>
                        </div>
                    </div>
                    <div class="mb-3">
                        <label class="form-label">Phone</label>
                        <input type="text" name="phone" class="form-control" required>
                    </div>
                    <button type="submit" name="register" class="btn btn-primary w-100">
                        Register
                    </button>

                </form>

            </div>

        </div>
    </div>
</div>
<script> // making password visible
function togglePassword(id) {
    var input = document.getElementById(id);

    if (input.type === "password") {
        input.type = "text";
    } else {
        input.type = "password";
    }
}
</script>
```

login_select.php

```php
<?php include 'menu.php'; ?>
<button onclick="history.back()" class="btn btn-light border mb-3">
    ← Back
</button>
<div class="container text-center mt-5">
    <h2>Clinic Booking System</h2>
    <p>Please select login type</p>
    <a href="admin_login.php" class="btn btn-danger btn-lg m-3">Admin Login</a>
    <a href="client_login.php" class="btn btn-primary btn-lg m-3">Client Login</a>
    <p class="mt-4">
        Haven't created an account yet?
        <a href="register.php">Register here</a>
    </p>
</div>
```

login.php

```php
<?php
include 'menu.php';
session_start();
include 'db_connect.php';

if (isset($_POST['login'])) {
    // getting input
    $email = $_POST['email'];
    $password = $_POST['password'];
    // retriving data input
    $sql = "SELECT * FROM users WHERE email='$email'";
    $result = mysqli_query($conn, $sql);
    $row = mysqli_fetch_assoc($result);
    //varify
    if ($row && password_verify($password, $row['password'])) {
        $_SESSION['email'] = $email;
        $_SESSION['role'] = $row['role'];
       // directing role
        if ($row['role'] == 'admin') {
            header("Location: admin_dashboard.php");
        } else {
            header("Location: patient_dashboard.php");
        }
        exit();
    } else {
        echo "<div class='alert alert-danger'>Invalid Email or Password</div>";
    }
}
?>

<form method="POST">
    Email: <input type="email" name="email" required><br><br>
    Password: <input type="password" name="password" required><br><br>
    <button type="submit" name="login">Login</button>
</form>
```

admin_login.php

```php
<?php
session_start();
include 'menu.php';
include 'db_connect.php';

$error = "";
if (isset($_POST['login'])) {
    $email = $_POST['email'];
    $password = $_POST['password'];
    $sql = "SELECT * FROM users WHERE email='$email' AND role='admin'";
    $result = mysqli_query($conn, $sql);
    $row = mysqli_fetch_assoc($result);
    if ($row) {
        // support hash pw and plain one
        if (password_verify($password, $row['password']) || $password == $row['password']) {
            $_SESSION['email'] = $email;
            $_SESSION['role'] = 'admin';
            header("Location: admin_dashboard.php");
            exit();
        } else {
            $error = "Invalid Admin Login";
        }
    } else {
        $error = "Admin not found";
    }
}
```

```php
<div class="container mt-5">
    <div class="row justify-content-center">
        <div class="col-md-5">

            <div class="card shadow p-4">

                <h3 class="text-center mb-4">🌓 Admin Login</h3>
                <?php if ($error) { ?>
                    <div class="alert alert-danger"><?php echo $error; ?></div>
                <?php } ?>

                <form method="POST">
                    <div class="mb-3">
                        <label class="form-label">Admin Email</label>
                        <input type="email" name="email" class="form-control" required>
                    </div>
                    <div class="mb-3">
                        <label class="form-label">Password</label>

                        <div class="input-group">
                            <input type="password" name="password" id="password" class="form-control" required>

                            <button type="button" class="btn btn-outline-secondary" onclick="togglePassword()">
                                👁
                            </button>
                        </div>
                    </div>
                    <button type="submit" name="login" class="btn btn-danger w-100">
                        Login as Admin
                    </button>
                </form>
            </div>
        </div>
    </div>
</div>

<script>
function togglePassword() {
    var input = document.getElementById("password");
    if (input.type === "password") {
        input.type = "text";
    } else {
        input.type = "password";
    }
}
</script>
<button onclick="history.back()" class="btn btn-light border mb-3">
    ⬅ Back
</button>
```

client_login.php

```php
<?php
session_start();
include 'menu.php';
include 'db_connect.php';

if (isset($_POST['login'])) {
    $email = $_POST['email'];
    $password = $_POST['password'];
    // getting gmail data
    $sql = "SELECT * FROM users WHERE email='$email' AND role='patient'";
    $result = mysqli_query($conn, $sql);
    $row = mysqli_fetch_assoc($result);
    // password verification
    if ($row && password_verify($password, $row['password'])) {

        $_SESSION['email'] = $email;
        $_SESSION['role'] = 'patient';
        header("Location: patient_dashboard.php");
        exit();
    } else {
        echo "<div class='alert alert-danger'>Invalid Client Login</div>";
    }
}
?>
```

```php
<div class="container mt-5">
    <div class="row justify-content-center">
        <div class="col-md-5">
            <div class="card shadow p-4">
                <h3 class="text-center mb-4">🔐 Client Login</h3>
                <form method="POST">
                    <div class="mb-3">
                        <label class="form-label">Email</label>
                        <input type="email" name="email" class="form-control" required>
                    </div>
                    <div class="mb-3">
                        <label class="form-label">Password</label>
                        <div class="input-group">
                            <input type="password" name="password" id="password" class="form-control" required>
                            <button type="button" class="btn btn-outline-secondary" onclick="togglePassword()">
                                👁
                            </button>
                        </div>
                    </div>
                    <div class="mb-3 text-end">
                        <a href="forgot_password.php">Forgot Password?</a>
                    </div>
                    <button type="submit" name="login" class="btn btn-primary w-100">
                        Login
                    </button>
                </form>
            </div>
        </div>
    </div>
</div>

<script>
function togglePassword() {
    var input = document.getElementById("password");
    if (input.type === "password") {
        input.type = "text";
    } else {
        input.type = "password";
    }
}
</script>
<button onclick="history.back()" class="btn btn-light border mb-3">
    ⬅ Back
</button>
```

admin_dashboard.php

```php
<?php
session_start();

if ($_SESSION['role'] != 'admin') {  // checking role
    header("Location: login.php");
    exit();
}
include 'menu.php';
?>

<div class="container mt-5">
    <h2 class="text-center mb-3">Admin Dashboard</h2>
    <p class="text-center text-muted mb-5">
        Welcome Admin. Manage clinic operations below.
    </p>
    <div class="row g-4">
        <div class="col-md-4">
            <div class="card shadow text-center p-4">
                <h4>📅 Appointments</h4>
                <p>Manage all bookings</p>
                <a href="view_appointments.php" class="btn btn-primary">Manage</a>
            </div>
        </div>
        <div class="col-md-4">
            <div class="card shadow text-center p-4">
                <h4>👨‍⚕️ Doctors</h4>
                <p>View doctor list</p>
                <a href="doctor_list.php" class="btn btn-success">View</a>
            </div>
        </div>
        <div class="col-md-4">
            <div class="card shadow text-center p-4">
                <h4>➕ Add Doctor</h4>
                <p>Add new doctor</p>
                <a href="add_doctor.php" class="btn btn-warning">Add</a>
            </div>
        </div>
        <div class="col-md-4">
            <div class="card shadow text-center p-4">
                <h4>📆 Today</h4>
                <p>View today's schedule</p>
                <a href="today_schedule.php" class="btn btn-info">View</a>
            </div>
        </div>
        <div class="col-md-4">
            <div class="card shadow text-center p-4">
                <h4>🚪 Logout</h4>
                <p>Sign out safely</p>
                <a href="logout.php" class="btn btn-danger">Logout</a>
            </div>
        </div>
    </div>
</div>
```

patient_dashboard.php

```php
<?php
session_start();

if ($_SESSION['role'] != 'patient') // check role and direct page
{
    header("Location: login_select.php");
    exit();
}
include 'menu.php';
?>
<div class="container mt-5">
    <h2 class="text-center mb-4">Patient Dashboard</h2>
    <div class="row text-center">
        <div class="col-md-4">
            <div class="card p-4 shadow">
                <h4>Book Appointment</h4>
                <a href="doctor_list.php" class="btn btn-primary">Book Now</a>
            </div>
        </div>
        <div class="col-md-4">
            <div class="card p-4 shadow">
                <h4>My Appointments</h4>
                <a href="my_appointments.php" class="btn btn-success">View</a>
            </div>
        </div>
        <div class="col-md-4">
            <div class="card p-4 shadow">
                <h4>Logout</h4>
                <a href="logout.php" class="btn btn-danger">Logout</a>
            </div>
        </div>
    </div>
</div>

```

forgot_password.php

```php
<?php include 'menu.php'; include 'db_connect.php'; ?>

<div class="container mt-5">
    <div class="row justify-content-center">
        <div class="col-md-5">
            <div class="card p-4 shadow">
                <h4 class="text-center mb-3">🔒 Reset Password</h4>
                <form method="POST">
                    <div class="mb-3">
                        <label>Email</label>
                        <input type="email" name="email" class="form-control" required>
                    </div>
                    <div class="mb-3">
                        <label>New Password</label>
                        <input type="password" name="password" class="form-control" required>
                    </div>
                    <button class="btn btn-primary w-100" name="reset">
                        Reset Password
                    </button>
                </form>
                <?php
                if (isset($_POST['reset'])) {
                    $email = $_POST['email'];
                    if (isset($_POST['reset'])) {
$email = $_POST['email'];
$raw_password = $_POST['password'];

// check pw valid
if (!preg_match("/^(?=.*[A-Z]).{6,}$/", $raw_password)) {

    echo "<div class='alert alert-danger mt-3'> Password must be at least 6 characters and contain at least one uppercase letter. </div>";
} else {
    $password = password_hash($raw_password, PASSWORD_DEFAULT); // hasing pw after valid
    $check = mysqli_query($conn, "SELECT * FROM users WHERE email='$email'");
    if (mysqli_num_rows($check) > 0) {
        mysqli_query($conn, "UPDATE users
                             SET password='$password'
                             WHERE email='$email'");
        echo "<div class='alert alert-success mt-3'> Password updated successfully! </div>";
    } else {
        echo "<div class='alert alert-danger mt-3'> Email not found! </div>";
    }
}
}
 $check = mysqli_query($conn, "SELECT * FROM users WHERE email='$email'");
 if (mysqli_num_rows($check) > 0) {
 mysqli_query($conn, "UPDATE users
 SET password='$password'
 WHERE email='$email'");
 echo "<div class='alert alert-success mt-3'> Password updated successfully! </div>";
  }
 else {
 echo "<div class='alert alert-danger mt-3'> Email not found! </div>";
}
}
?>
</div>
</div>
</div>
</div>
```

add_doctor.php

```php
<?php
session_start();

if ($_SESSION['role'] != 'admin') {
    header("Location: login.php");
    exit();
}

include 'menu.php';
include 'db_connect.php';

if (isset($_POST['add'])) {
    // inputing doc data
    $name = $_POST['name'];
    $spec = $_POST['spec'];
    $day = $_POST['day'];
    $start = $_POST['start_time'];
    $end = $_POST['end_time'];
    mysqli_query($conn, "INSERT INTO doctors
        (doctor_name, specialization, available_day, start_time, end_time)
        VALUES ('$name', '$spec', '$day', '$start', '$end')");
    echo "<div class='container mt-3'><div class='alert alert-success'> Doctor added successfully! </div></div>";
}
?>
<div class="container mt-4">
<h2>Add Doctor</h2>
<form method="POST">
    <div class="mb-3">
        <label class="form-label">Doctor Name</label>
        <input type="text" name="name" class="form-control" required>
    </div>
    <div class="mb-3">
        <label class="form-label">Specialization</label>
        <select name="spec" class="form-select" required>
            <option value="">-- Select Specialization --</option>
            <option value="General">🩺 General Practice</option>
            <option value="Cardiology">❤️ Cardiology</option>
            <option value="Eye">👁️ Eye Care</option>
            <option value="Dental">🦷 Dental</option>
            <option value="Women">🎗️ Women's Health</option>
            <option value="Vaccination">💉 Vaccination</option>
        </select>
    </div>
    <div class="mb-3">
        <label class="form-label">Available Day</label>
        <input type="text" name="day" class="form-control" placeholder="e.g. Monday - Friday" required>
    </div>
    <div class="mb-3">
        <label class="form-label">Start Time</label>
        <input type="time" name="start_time" class="form-control" required>
    </div>
    <div class="mb-3">
        <label class="form-label">End Time</label>
        <input type="time" name="end_time" class="form-control" required>
    </div>
    <button type="submit" name="add" class="btn btn-primary">
        Add Doctor
    </button>
</form>
</div>
<button onclick="history.back()" class="btn btn-light border mb-3">
    ⬅ Back
</button>
```

edit_doctor.php

```php
<?php
session_start();

if ($_SESSION['role'] != 'admin') {
    header("Location: login.php");
    exit();
}
include 'menu.php';
include 'db_connect.php';
if (!isset($_GET['id'])) {
    echo "No doctor selected.";   // checking doc selection
    exit();
}
$id = $_GET['id'];
$result = mysqli_query($conn, "SELECT * FROM doctors WHERE doctor_id='$id'"); // retrieve doc data from db to check
$row = mysqli_fetch_assoc($result);
if (!$row) {
    echo "Doctor not found.";
    exit();
}
if (isset($_POST['update'])) {
    $name = $_POST['name'];
    $spec = $_POST['spec'];
    $day = $_POST['day'];
    $start = $_POST['start_time'];
    $end = $_POST['end_time'];
    mysqli_query($conn, "UPDATE doctors
        SET doctor_name='$name',
            specialization='$spec',
            available_day='$day',
            start_time='$start',
            end_time='$end'
        WHERE doctor_id='$id'");
    echo "<div class='container mt-3'><div class='alert alert-success'> Doctor updated successfully! </div></div>";
}
?>
```

```php
<div class="container mt-4">
<h2>Edit Doctor</h2>

<form method="POST">
    <div class="mb-3">
        <label class="form-label">Doctor Name</label>
        <input type="text" name="name" class="form-control"
               value="<?php echo $row['doctor_name']; ?>" required>
    </div>
    <div class="mb-3">
        <label class="form-label">Specialization</label>
        <select name="spec" class="form-select" required>
            <option value="General" <?php if($row['specialization']=="General") echo "selected"; ?>>
                🩺 General Practice
            </option>
            <option value="Cardiology" <?php if($row['specialization']=="Cardiology") echo "selected"; ?>>
                ❤️ Cardiology
            </option>
            <option value="Eye" <?php if($row['specialization']=="Eye") echo "selected"; ?>>
                👁️ Eye Care
            </option>
            <option value="Dental" <?php if($row['specialization']=="Dental") echo "selected"; ?>>
                🦷 Dental
            </option>
            <option value="Women" <?php if($row['specialization']=="Women") echo "selected"; ?>>
                🎗️ Women's Health
            </option>
            <option value="Vaccination" <?php if($row['specialization']=="Vaccination") echo "selected"; ?>>
                💉 Vaccination
            </option>
        </select>
    </div>
    <div class="mb-3">
        <label class="form-label">Available Day</label>
        <input type="text" name="day" class="form-control"
               value="<?php echo $row['available_day']; ?>" required>
    </div>
    <div class="mb-3">
        <label class="form-label">Start Time</label>
        <input type="time" name="start_time" class="form-control"
               value="<?php echo $row['start_time']; ?>" required>
    </div>
    <div class="mb-3">
        <label class="form-label">End Time</label>
        <input type="time" name="end_time" class="form-control"
               value="<?php echo $row['end_time']; ?>" required>
    </div>
    <button type="submit" name="update" class="btn btn-success">
        Update Doctor
    </button>
</form>
</div>
<button onclick="history.back()" class="btn btn-light border mb-3">
    ← Back
</button>
```

delete_doctor.php

```php
<?php
session_start();
include 'db_connect.php';

if (!isset($_GET['id'])) {
    echo "No doctor selected.";
    exit();
}
$id = $_GET['id'];
mysqli_query($conn, "DELETE FROM appointments WHERE doctor_id='$id'");
mysqli_query($conn, "DELETE FROM doctors WHERE doctor_id='$id'");
header("Location: doctor_list.php");
exit();
?>
```

db_connect.php

```php
<?php
$conn = mysqli_connect("localhost", "root", "", "clinic_booking_db");
if (!$conn) {
    die("Connection failed: " . mysqli_connect_error());
}
?>
```

doctor_list.php

```php
<?php
session_start();
include 'menu.php';
include 'db_connect.php';

$sql = "SELECT * FROM doctors ORDER BY specialization"; // taking all doc data
$result = mysqli_query($conn, $sql);

$current_spec = "";
?>
<div class="container mt-4">
<h2>Choose Doctor</h2>
<?php while($row = mysqli_fetch_assoc($result)) { ?>
    <?php if ($current_spec != $row['specialization']) // display spec
    {
        $current_spec = $row['specialization'];
    ?>
        <h4 class="mt-4">
            <?php // adding icon based on specialization
            if ($current_spec == "Cardiology") echo "❤️ ";
            elseif ($current_spec == "Eye") echo "👁️ ";
            elseif ($current_spec == "Dental") echo "🦷 ";
            elseif ($current_spec == "Women") echo "🤰 ";
            elseif ($current_spec == "Vaccination") echo "💉 ";
            else echo "🩺 ";
            ?>
            <?php echo $current_spec; ?>
        </h4>
        <hr>
    <?php } ?>
    <div class="card mb-3 p-3 shadow-sm">
        <strong><?php echo $row['doctor_name']; ?></strong><br>
        Day: <?php echo $row['available_day']; ?><br>
        Time: <?php
                echo date("h:i A", strtotime($row['start_time'])) .
                    " - " .
                    date("h:i A", strtotime($row['end_time']));
                ?>
        <?php if(isset($_SESSION['role']) && $_SESSION['role'] == 'admin') // checking role
        { ?>
            <a href="edit_doctor.php?id=<?php echo $row['doctor_id']; ?>" class="btn btn-warning btn-sm">Edit</a>
            <a href="delete_doctor.php?id=<?php echo $row['doctor_id']; ?>"
               class="btn btn-danger btn-sm"
               onclick="return confirm('Delete this doctor?')">Delete</a>
        <?php } else { ?>
            <a href="book_appointment.php?doctor_id=<?php echo $row['doctor_id']; ?>"
               class="btn btn-primary btn-sm">Book Appointment</a>
        <?php } ?>
    </div>
<?php } ?>
</div>
<button onclick="history.back()" class="btn btn-light border mb-3">
    ← Back
</button>
```

view_appointments/php

```php
<?php
session_start();
include 'menu.php';
include 'db_connect.php';

// admin check
if ($_SESSION['role'] != 'admin') {
    header("Location: login.php");
    exit();
}
// get patient and doc info
$sql = "SELECT appointments.*, users.name, doctors.doctor_name
        FROM appointments
        JOIN users ON appointments.user_id = users.user_id
        JOIN doctors ON appointments.doctor_id = doctors.doctor_id";
$result = mysqli_query($conn, $sql);
// check sql error
if (!$result) {
    die("Query failed: " . mysqli_error($conn));
}
?>
<div class="container mt-4">
    <button onclick="history.back()" class="btn btn-light border mb-3">
        ← Back
    </button>
    <h2 class="mb-4">Appointment List</h2>
    <table class="table table-bordered table-hover shadow-sm text-center">
        <thead class="table-dark">
            <tr>
                <th>Patient Name</th>
                <th>Doctor</th>
                <th>Date</th>
                <th>Time</th>
                <th>Status</th>
                <th>Action</th>
            </tr>
        </thead>
        <tbody>
```

```php
        <?php while($row = mysqli_fetch_assoc($result)) { ?>
            <tr>
                <td><?php echo $row['name']; ?></td>
                <td><?php echo $row['doctor_name']; ?></td>
                <td><?php echo $row['appointment_date']; ?></td>
                <td><?php echo $row['appointment_time']; ?></td>
                <td>
                    <?php if($row['status'] == 'Approved') { ?>
                        <span class="badge bg-success">Approved</span>
                    <?php } elseif($row['status'] == 'Cancelled') { ?>
                        <span class="badge bg-danger">Cancelled</span>
                    <?php } else { ?>
                        <span class="badge bg-warning text-dark">Pending</span>
                    <?php } ?>
                </td>
                <td>
                    <?php if ($row['status'] == 'Pending') { ?> // checking pedning or not

                        <a href="update_status.php?id=<?php echo $row['appointment_id']; ?>&status=Approved"
                          class="btn btn-success btn-sm">
                          Approve
                        </a>
                        <a href="update_status.php?id=<?php echo $row['appointment_id']; ?>&status=Cancelled"
                          class="btn btn-danger btn-sm"
                          onclick="return confirm('Cancel this appointment?')">
                          Cancel
                        </a>
                    <?php } else { ?>
                        <button class="btn btn-secondary btn-sm" disabled>
                            No Action
                        </button>
                    <?php } ?>
                </td>
            </tr>
        <?php } ?>
        </tbody>
    </table>
</div>
```

update_status.php

```php
<?php
include 'db_connect.php';

$id = $_GET['id'];
$status = $_GET['status'];
// checking current status
$check = mysqli_query($conn, "SELECT status FROM appointments WHERE appointment_id='$id'");
$row = mysqli_fetch_assoc($check);
// if not pedning, cannot make action
if ($row['status'] != 'Pending') {
    echo "<script>
            alert('Action not allowed! Appointment already processed.');
            window.location='view_appointments.php';
          </script>";
    exit();
}
// allow action if pedning
mysqli_query($conn, "UPDATE appointments SET status='$status' WHERE appointment_id='$id'");
header("Location: view_appointments.php");
exit();
?>
```

today_schedule.php

```php
<?php
session_start();
include 'menu.php';
include 'db_connect.php';

if ($_SESSION['role'] != 'admin') {
    header("Location: login_select.php");
    exit();
}
$today = date('Y-m-d');
$sql = "SELECT appointments.*, users.name, doctors.doctor_name
        FROM appointments
        JOIN users ON appointments.user_id = users.user_id
        JOIN doctors ON appointments.doctor_id = doctors.doctor_id
        WHERE appointment_date = '$today'
        AND status = 'Approved'
        ORDER BY appointment_time";
$result = mysqli_query($conn, $sql);
?>
<div class="container mt-4">
<h2>Today's Approved Appointments</h2>
<table class="table table-bordered">
<tr>
    <th>Patient</th>
    <th>Doctor</th>
    <th>Date</th>
    <th>Time</th>
</tr>
<?php while($row = mysqli_fetch_assoc($result)) { ?>
<tr>
    <td><?php echo $row['name']; ?></td>
    <td><?php echo $row['doctor_name']; ?></td>
    <td><?php echo $row['appointment_date']; ?></td>
    <td><?php echo $row['appointment_time']; ?></td>
</tr>
<?php } ?>
</table>
</div>
<button onclick="history.back()" class="btn btn-light border mb-3">
    ← Back
</button>

```

my_appointments.php

```php
<?php
session_start();
include 'menu.php';
include 'db_connect.php';

if (!isset($_SESSION['email']) || $_SESSION['role'] != 'patient') // check valid for patient
{
    header("Location: login_select.php");
    exit();
}
$user_email = $_SESSION['email'];
// user safe
$user_query = mysqli_query($conn, "SELECT * FROM users WHERE email='$user_email'");
$user = mysqli_fetch_assoc($user_query);

// prevent nuill
if (!$user) {
    echo "<div class='alert alert-danger'>User not found. Please login again.</div>";
    exit();
}
$user_id = $user['user_id'];
// retrieve doc data that already make appointment
$sql = "SELECT appointments.*, doctors.doctor_name
        FROM appointments
        JOIN doctors ON appointments.doctor_id = doctors.doctor_id
        WHERE appointments.user_id = '$user_id'";
$result = mysqli_query($conn, $sql);
?>
<div class="container mt-4">
<h2>My Appointments</h2>
<table class="table table-bordered">
<tr>
    <th>Doctor</th>
    <th>Date</th>
    <th>Time</th>
    <th>Status</th>
    <th>Action</th>
</tr>
<?php while($row = mysqli_fetch_assoc($result)) { ?>
<tr>
    <td><?php echo $row['doctor_name']; ?></td>
    <td><?php echo $row['appointment_date']; ?></td>
    <td><?php echo $row['appointment_time']; ?></td>
    <td><?php echo $row['status']; ?></td>
    <td>
    <?php if($row['status'] != 'Cancelled') { ?>
        <a href="cancel_my_appointment.php?id=<?php echo $row['appointment_id']; ?>"
          onclick="return confirm('Cancel this appointment?')">
          Cancel
        </a>
    <?php } ?>
    </td>
</tr>
<?php } ?>
</table>
</div>
<button onclick="history.back()" class="btn btn-light border mb-3">
    ← Back
</button>
```

logout.php

```php
<?php
session_start();
session_unset();
session_destroy();

header("Location: index.php");
exit();
?>

```

create.php

```php
<?php include 'menu.php'; ?>
<div class="container mt-4">
    <h2>Contact Us</h2>
    <p>Email: clinic@gmail.com</p>
    <p>Phone: +60 123 456 789</p>
    <p>Address: Kuala Lumpur, Malaysia</p>
</div>

```

book_appointments.php

```php
<?php
include 'menu.php';
session_start();
include 'db_connect.php';

if (!isset($_SESSION['email'])) // check validation
{
    header("Location: login_select.php");
    exit();
}
$doctor_id = $_GET['doctor_id'];
$user_email = $_SESSION['email'];
// getting user data
$user_query = mysqli_query($conn, "SELECT * FROM users WHERE email='$user_email'");
$user = mysqli_fetch_assoc($user_query);
$user_id = $user['user_id'];

// getting doc data
$doc = mysqli_fetch_assoc(mysqli_query($conn,
    "SELECT * FROM doctors WHERE doctor_id='$doctor_id'"
));
if (!$doc) {
    die("Doctor not found.");
}
$start_time = strtotime($doc['start_time']);
$end_time = strtotime($doc['end_time']);
$message = "";
```

```php
if (isset($_POST['book'])) {

    $date = $_POST['date'];
    $time = $_POST['time'];

    // check double booking
    $check = mysqli_query($conn, "
        SELECT * FROM appointments
        WHERE doctor_id='$doctor_id'
        AND appointment_date='$date'
        AND appointment_time='$time'
        AND status != 'Cancelled'
    ");

    if (mysqli_num_rows($check) > 0) {
        $message = "<div class='alert alert-danger'> This time slot is already booked. Please choose another time. </div>";
    } else {
        $sql = "INSERT INTO appointments
                (user_id, doctor_id, appointment_date, appointment_time, status)
                VALUES ('$user_id', '$doctor_id', '$date', '$time', 'Pending')";
        if (mysqli_query($conn, $sql)) {
            $message = "<div class='alert alert-success'> Appointment Booked Successfully! </div>";
        }
    }
}
?>
```

```php
<div class="container mt-5">
    <button onclick="history.back()" class="btn btn-light border mb-3">
        ← Back
    </button>
    <div class="row justify-content-center">
        <div class="col-md-6">
            <div class="card shadow p-4">
                <h3 class="text-center mb-4">📅 Book Appointment</h3>
                <?php echo $message; ?>
                <form method="POST">
                    <div class="mb-3">
                        <label class="form-label">Appointment Date</label>
                        <input type="date" name="date" class="form-control" required>
                    </div>
                    <div class="mb-3">
                        <label class="form-label">Appointment Time</label>
                        <select name="time" class="form-select" required>
                            <option value="">-- Select Time --</option>
                            <?php
                            $current = $start_time;
                            while ($current < $end_time) {
                                $time = date("h:i A", $current);
                                echo "<option>$time</option>";
                                $current = strtotime("+1 hour", $current);
                            }
                            ?>
                        </select>
                    </div>
                    <button type="submit" name="book"
                            class="btn btn-primary w-100">
                        Book Appointment
                    </button>
                </form>
            </div>
        </div>
    </div>
</div>
```

cancel_appointment.php

```php
<?php
session_start();
include 'db_connect.php';
$id = $_GET['id'];
mysqli_query($conn, "UPDATE appointments
                     SET status='Cancelled'
                     WHERE appointment_id='$id'");

header("Location: my_appointments.php");
exit();
?>

```

cancel_my_appointment.php

```php
<?php
session_start();
include 'db_connect.php';

if ($_SESSION['role'] != 'patient') {
    header("Location: login_select.php");
    exit();
}
$id = $_GET['id'];
mysqli_query($conn, "UPDATE appointments
                     SET status='Cancelled'
                     WHERE appointment_id='$id'");

header("Location: my_appointments.php");
exit();
?>

```